Tip of the Iceberg: Tip Reporting at U.S. Restaurants, 2005-2018

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Abstract

Tipping is a significant form of compensation for many restaurant jobs, but it is poorly measured and therefore not well understood. We combine several large administrative and survey datasets and document patterns in tip reporting that are consistent with systematic under-reporting of tip income. Our analysis indicates that although the vast majority of tipped workers do report earning some tips, the dollar value of tips is under-reported and is sensitive to reporting incentives. In total, we estimate that about eight billion in tips paid at full-service, single-location, restaurants were not captured in tax data annually over the period 2005-2018. Due to changes in payment methods and reporting incentives, tip reporting has increased over time. Our findings have implications for downstream measures dependent on accurate measures of compensation including poverty measurement among tipped restaurant workers.

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1 Introduction

Tipping is ubiquitous. Nearly half of jobs at U.S. full-service restaurants reported tip earnings to the Internal Revenue Service (IRS) over the period 2005–2018. Total reported tips averaged $24.2 billion per year over this period. Despite the prevalence of tipping, because tips represent a direct transaction between customers and workers, little is known about how well tips are reported in administrative data. In this paper, we analyze tax reports at both the individual level and the employer level to assess the reliability of tip reporting. We document that the vast majority of tipped jobs report receiving at least some tips, allowing us to identify these workers in tax reports. At the same time, the evidence is consistent with substantial under-reporting of tips received. The dollar value of tips reported is sensitive to a variety of incentives that apply differentially by employer size, location, and time.

The two most commonly used sources of data on workers' earnings are administrative records, which include tax filings, and responses to household surveys. Recent research comparing earnings measures from different sources has highlighted that neither tax reports nor survey responses give a complete measure of earnings as both sources have the potential for under-reporting (see, e.g., Abowd and Stinson, 2013). Administrative data may be particularly problematic for tipped jobs because of the informal nature of this form of compensation. Although an individual worker's tips may be under-reported or over-reported, most observers believe that tips are, on the whole, under-reported (see, e.g., Treasury Inspector General for Tax Administration, 2018). The IRS estimates that only about 45% of tips are reported on tax filings (Internal Revenue Service; Research, Applied Analytics and Statistics, 2022, pp. 2, 18), but this number is based on the assumption that tips are under-reported at the same rate as self-employment income, not on studies specifically targeting tipped workers.

To fill the gap in our understanding of the quality of administrative earnings data for tipped workers, we combine several large administrative and survey datasets on both restaurants and workers, covering U.S. single-unit (SU) full-service (FS) restaurants from 2005 through 2018. The FS restaurant industry is defined as the set of "establishments primarily

engaged in providing food services to patrons who order and are served while seated (i.e., waiter/waitress services) and pay after eating."[1] Published data from the Economic Census show the U.S. had approximately 250,000 FS restaurants in 2017, about 210,000 of them operated by SU firms.[2] We focus on SU restaurants because we have higher-frequency and higher-quality data for these restaurants. Most importantly, we are able to match jobs to exact work locations.

Restaurants are a natural setting to study tipping for several reasons. First, FS restaurants account for the lion's share of tips reported annually to the IRS: tips reported by FS restaurant jobs account for over 80% of total reported tips economy-wide.[3] Second, tips comprise a large share of employee compensation at restaurants. At the restaurant level, reported tips accounted for more than 20% of employee compensation at FS restaurants, or about $90,000 per restaurant per year, compared to $330,000 in payroll for all workers. Third, existing laws and regulations introduce variation in incentives that provide identification for our analysis. Because tips are so prevalent and most restaurant workers are paid hourly, there is widespread use of the tipped-subminimum wage, a lower hourly rate permitted for tipped jobs in most states. Employers are required to ensure that workers paid the subminimum wage earn enough in tips to make up the difference between the subminimum wage and the standard minimum wage. If tips are too low to cover the gap, the employer is responsible for making up the deficit. In practice, this forces the formal reporting of at least some tips for many jobs to document minimum-wage compliance.

The evidence we find suggests that most workers who earn tips report at least some tips on their tax returns, especially after 2010. For example, from 2011 to 2018, the share of jobs reporting some tips to the IRS is almost identical – 49.3% versus 49.5% – in states with

[1]Source: Executive Office of the President, Office of Management and Budget (2022, p. 563).

[2]See `https://data.census.gov/table?text=7225&g=010XX00US&y=2017&n=722511:722513&tid=ECNSIZE2017.EC1700SIZESUMU`, accessed November 11, 2023.

[3]Total tips reported on Form W-2 from 2008–2018 averaged about $30 billion per year (Internal Revenue Service, 2017).

and without legal subminimum wages that generate additional reporting incentives. We also validate workers' tip reporting using information on worker occupation, which is available for a subset of workers from household surveys. We find that the vast majority of servers report tips on their Form W-2, whereas restaurant workers in other occupations (such as dishwashers) are much less likely to report tips. Finally, we show that the share of jobs reporting tips to the IRS is higher than the share reporting tips on household surveys.

We are less confident about *restaurant-level* reporting. One indication of a problem is that more than a quarter of SU FS restaurants report no tips. In addition, tip-reporting rates are lower in states without subminimum wages than in states with a subminimum wage, consistent with strategic reporting behavior by restaurants. Among restaurants that do report tips earned by their workers, the ratio of reported tips to revenue is low compared to tipping norms in the United States. Finally, we exploit an IRS rule that requires many restaurants to report tips equal to at least 8% of revenue and show bunching at that threshold for restaurants to which the rule applies. Taken together, these results point to substantial under-reporting of tips.

The under-reporting of tips at the restaurant level is reflected also in job-level reported earnings. We find that workers in jobs subject to a two-tier minimum wage bunch their reported tips at exactly the level needed to meet minimum-wage requirements.

We do a back-of-the-envelope calculation of missing tips by assuming that 95% of restaurant revenue comes from food and beverage sales and that customers tip on average 15% of the cost of their meals. Our calculations imply that an average of about eight billion dollars in tips may have gone unreported annually by SU FS restaurants over the period of our study. Of this amount, we attribte about a quarter to restaurants that reported zero tips, and the remaining three quarters to restaurants that reported some, but not all, of the tips their workers earned. Our estimates assume that approximately half of the jobs at restaurants with no reported tips earn (but do not report) some tips. Among jobs with reported tips, we estimate that, on average, 60% of tips are reported and show that tip reporting has

increased over time. In short, while IRS estimates suggest that the majority of the iceberg is not visible (i.e., not measured), we find that more of the tip iceberg is actually visible (i.e., measured). Finally, we use data from 2018 to illustrate how including or excluding tip income from total earnings affects poverty measurement.

The rest of this paper is organized as follows. Section 2 details the institutional and legal environment that motivates our empirical exercises. Section 3 describes our data sources, with some omitted details presented in Appendix A. We discuss workers' tip reporting and its responsiveness to reporting incentives in Section 4. In Section 5, we turn to incentives that apply to the restaurant, and examine how these incentives interact with job-level reporting incentives. In Section 6 we combine our findings on reported tips with assumptions about actual tipping rates to deduce the likely level of tips missing in the data and the potential change in the poverty rate for tipped restaurant workers if these missing tips were reported as income. Section 7 concludes with implications for both policy and income measurement.

2 Institutional and Legal Setting

Understanding actual tip earnings is impossible without understanding how legal and institutional incentives shape reporting of this type of compensation. A tip, or gratuity, is a direct payment from a customer to a worker. This payment bypasses the employer but represents job-related earnings for the worker. Consequently, the employer may not have direct knowledge of whether or how much workers receive in tips.[4] The employer is required to track worker tips, but this tracking often relies on workers' reports of their tip earnings. In addition, the tax liability on tip income is different for workers and employers. Workers owe Federal Insurance Contributions Act (FICA) payroll taxes (7.65%) on all wages and

[4]An exception, increasingly common, is when tips are left on a charge card. In that case the employer processes the charge but is required under federal law to dispense the funds to the worker no later than the regular payday, whether or not the employer has received the funds from the credit- or debit-card company. See `https://www.dol.gov/agencies/whd/fact-sheets/15-tipped-employees-flsa`, last accessed April 30, 2023.

tips. Employers owe the same percentage but can deduct their share of FICA payroll taxes on "excess" tips from any business income taxes they owe, where excess tips are defined as that portion above what is needed for the worker to earn $5.15 an hour.[5]

Combined, these features of tips imply that, first, tips are often poorly documented, and, second, workers have an incentive to minimize their tip reporting because of the tax implications of reporting more tips.[6] In contrast, from the employer's perspective, the disincentive to report tips is lower because the employer's tax liability generally does not increase with reported tips.

In most U.S. states, minimum-wage policy also has implications for tip reporting.[7] During the period we study, workers in tipped jobs in 43 states and the District of Columbia were subject to lower, subminimum, wages, if they earned at least $30 a month in tips.[8] The difference between the standard minimum wage and the tipped-subminimum wage is called the "tip credit"; it is a credit from the point of view of employers. Employers are required to ensure that tipped workers' wages and tips sum to at least the full minimum wage. In many cases, the tip credit is substantial: In 19 states and the District of Columbia, the tip credit in 2018 exceeded 70% of the minimum wage. In these states, the hourly subminimum

[5]FICA taxes include Social Security taxes (6.2%) and Medicare taxes (1.45%). There is a cap on wages subject to Social Security taxes which varies by year. In 2018 it was $128,400. See `https://www.ssa.gov/oact/cola/cbb.html`, last accessed April 30, 2024. Although the federal minimum wage increased from $5.15 in July 2007, for the purpose of calculating employers' FICA tax liability, the threshold has remained frozen at $5.15. In practice, employers pay the full employer share of FICA taxes due on each worker's total earnings and receive a dollar-for-dollar credit that can be deducted from the restaurant's income tax, if any is owed; unused credit may be carried back one year or forward for 20 years. This system enables the Social Security Administration (SSA) to collect the same amount of FICA taxes for restaurant workers as for any other type of worker and for the workers to receive credit for their earnings in the Social Security system. Restaurant owners are subsidized by general income-tax revenue to help cover the cost of FICA taxes on tips. One of the main purposes of this system is to avoid a disincentive on the part of the restaurant to report tips (U.S. Department of Treasury, Office of Tax Analysis, 2016).

[6]Alm, Deskins, and McKee (2009) use experimental evidence to argue that tips, like self-employment income, are under-reported to the tax authority, and Anderson and Bodvarsson (2005) show using public, state-level data, that patterns of tip reporting are consistent with tax evasion, but neither paper provides a quantitative estimate of under-reporting.

[7]The 1966 amendment to the Fair Labor Standards Act simultaneously codified the use of tips to meet part of employer minimum-wage obligations and extended minimum-wage coverage to hotel and restaurant jobs (Eisenberg and Williams, 2015; Meyer, 2017).

[8]The exceptions are Alaska, California, Minnesota, Montana, Nevada, Oregon, and Washington.

wage for tipped workers is $2.13/hour; if a worker earning this subminimum reports less than $5.12/hour in tips, the employer is required to make up the difference. The two-tier minimum-wage system creates an incentive for many employers to ensure that workers report enough tips to cover the tip credit.[9]

Despite their very different minimum-wage laws, there is no evidence that patrons behave differently in states with and without a tip credit. For example, although travel and etiquette guides highlight differences in tipping norms between the U.S. and European countries, they do not single out states without a tip credit to suggest that patrons tip less there (Azar, 2020). In 2014, the payment service Square released data on state-level average tip rates for transactions that used its service. The average tip rate was 16.4% for transactions with tips in the seven states that had no tip credit, and 16.3% on average in the 17 states that used the federal minimum wage in 2014.

Restaurants also face a tip-reporting threshold. The Tax Equity and Fiscal Responsibility Act of 1982 requires restaurants report tips equal to at least 8% of revenue or face penalties (Robertson, Quinn, and Carr, 2006; Newman, 1988). An analysis of tax reports for 600 restaurants firms by the U.S. General Accounting Office found that these firms doubled their reporting of tips following the implementation of this revenue-share rule (U.S. General Accounting Office, 1986, p. 25).

These rules do not affect all restaurants equally. In many states, the very smallest businesses are exempt from minimum-wage laws. At the federal level, the 8% revenue-share rule for the ratio of tips to revenue only applies to restaurants with at least ten employees, collectively working a total of at least 80 hours, on a typical business day, and where tipping is customary.[10] In addition, the Fair Labor Standards Act (FLSA), which specifies the

[9]This is somewhat reminiscent of incentives to report self-employment income to meet Earned Income Tax Credit income thresholds (Imboden, Voorheis, and Weber, 2023). Unlike self-employment income, tip income is reported to the employer on a flow basis (See `https://www.irs.gov/businesses/small-businesses-self-employed/tip-recordkeeping-and-reporting`, accessed September 3, 3024). For this reason, we do not believe EITC thresholds are an important incentive for tip reporting.

[10]See `https://www.irs.gov/taxtopics/tc761`, accessed July 31, 2024.

federal minimum and submininum wages and some worker rights, exempts enterprises that have annual gross volume of sales of $500,000 or less and those with only one employee. However, workers handling credit-card transactions are deemed to be engaged in interstate commerce, which is covered under FLSA.[11] Consequently, the tip-reporting incentives for small restaurants are less clear than for larger ones, with FLSA coverage possibly varying by employee.

Tip reporting may have changed over time for several reasons. One reason is that, as a larger share of transactions use credit and debit cards, and more tips are included on those transactions, tracking of tips has improved.[12] This means that there is less need for "guesstimating" tips and a paper trail exists that may, in and of itself, encourage tip reporting. In addition, the FLSA minimum threshold has stayed at a fixed nominal value of $500,000 throughout the study period. As the real value of this threshold falls, the FLSA and its reporting incentives apply to a larger set of restaurants each year.[13]

In addition to these gradually increasing incentives, there may have been a discontinuity in restaurants' incentive to report tips in 2011. That year, a provision of the Housing and Economic Recovery Act went into effect, requiring that third-party network transactions, including tips, be reported to the IRS on Form 1099-K. Similar to the other reporting incentives, Form 1099-K is only required if the network processed a minimum number of

[11]For details, see `https://www.dol.gov/agencies/whd/fact-sheets/14-flsa-coverage` and `https://www.dol.gov/agencies/whd/fact-sheets/2-flsa-restaurants`, accessed April 24, 2024.

[12]The Survey of Consumer Payment Choice (SCPC) shows a decrease in the percentage of payments made in cash from about 30% in 2009 to under 25% by 2018. See `https://www.atlantafed.org/banking-and-payments/consumer-payments/survey-of-consumer-payment-choice`, accessed September 3, 2024. Gerdes, Hamburg, and Liu (2017) report that between 2000 and 2015, consumer non-cash payments grew by 4.5% per year, with the majority of that growth coming from increased use of debit cards.

[13]The threshold dates back to 1989 (Gitterman, 2013). Published EC figures show that, from 2007 to 2017, the share of FS restaurants operated by firms with nominal annual revenue below $500,000 fell from 44% to 36%, and the share of FS employment at these restaurants fell from 12% to 7%. Statistics derived from `https://data.census.gov/table/ECNSIZE2017.EC1700SIZEREVFIRM?text=7225&g=010XX00US&y=2017&n=722511` and `https://www2.census.gov/programs-surveys/economic-census/data/2007/sector72/EC0772SSSZ4.zip`, last accessed March 21, 2024.

transactions and a minimum value.[14] Form 1099-K is filed by the credit-card network and includes the gross amount processed through the network. This gross amount includes not only restaurant revenue but applicable sales taxes, service charges, tips, and any other charges. Research has shown that this type of information return can increase tax reporting. For example, Garin, Jackson, Koustas, and Miller (2023) show that self-employed workers who receive payments from online platforms such as ride-sharing apps are more likely to report self-employment income on their tax returns if they receive a Form 1099-K from the platform. Slemrod, Collins, Hoopes, Reck, and Sebastiani (2017) find an overall small impact of the introduction of Form 1099-K reporting, but a larger impact on a set of "at-risk" firms whose self-reported income is very close to the 1099-K amount. In our setting, although Form 1099-K does not separately report tips, it still may affect restaurants' incentives to report tips accurately to the IRS due to increased transparency, particularly for restaurants that accept credit and debit cards.

For employers, tips represent a means of compensating their employees that has fewer tax obligations and less risk than formal wage compensation. For workers, tip income is inherently riskier than wage earnings: some patrons tip well, and others do not leave a tip at all.[15] The requirement that the employer ensures the worker earns at least the statutory minimum wage is one way of mitigating that risk. Another is tip pooling, whereby tipped workers put some or all of their tips into a common pool from which they are redistributed. Mandatory tip pooling is allowed in most states and voluntary pooling is allowed in all states.[16] During most of the study period, mandatory tip pools could include only workers who interacted directly with customers and could not include managers (Meyer,

[14]From 2011 to 2018, the thresholds for a 1099-K were $20,000 in reportable transactions and more than 200 such transactions. See `https://www.irs.gov/pub/irs-prior/i1099k--2011.pdf` and `https://www.irs.gov/pub/irs-prior/i1099k--2018.pdf`, last accessed October 23, 2024. Some states have lower thresholds; for example, in Vermont starting in tax year 2017, the value threshold was reduced to $600 and the transaction count threshold was removed altogether (`https://tax.vermont.gov/news/1099-k-information-reporting`, last accessed October 23, 2024).

[15]Servers complain about the irregularity of tip income; see, e.g., Dublanica (2008); Woolery (2015).

[16]Eisenberg and Williams (2015) claim that most restaurants implement tip pooling (p. 394).

2017; Estreicher and Nash, 2018). (This last rule was relaxed in March 2018, near the end of our sample period; see Federal Register, 2021b, p. 52975.) Mandatory tip pooling may not eat into the tip credit and may not exceed a "customary and reasonable" percentage of an employee's tips, which the U.S. Department of Labor (DOL) defined as no more than 15% until 2011 (Eisenberg and Williams, 2015, p. 398; that threshold was relaxed following litigation). Tip pooling could increase reported tips by some workers (the net recipients) and reduce reported tips by other workers, but, as long as all the workers in the pool are tipped, it should not affect the selection of workers into tip reporting, and by construction cannot impact restaurant-level tip reporting.[17]

We end our sample in 2018 to capture the pre-pandemic, traditional tip-pooling equilibrium in the FS restaurant industry. The changes in tip-pooling policies in 2018, noted above, have the possibility of affecting not just how much workers earn in tips but also who gets tips. Focusing on the period before these changes took effect provides a cleaner setting. In addition, the Covid-19 pandemic had a profound impact on service industries in general and FS restaurants in particular, and changed the way many patrons tip. The diffusion of checkout systems that request tips across a wide array of businesses has elicited complaints about so-called "tipflation" in the post-pandemic period (e.g., Kelly, 2023), potentially further changing tipping behavior and reporting. Thus, our paper provides an important baseline for tip measurement prior to these dramatic changes.

Informed by our understanding of the institutional and legal environment, and the incentives it generates for reporting (or not reporting) tips, in Sections 4 and 5, we report the results of several exercises designed to assess how the quality of tip data reported to the IRS responds to these incentives, both at the job level and at the restaurant level.

[17]Informal sharing of tips, which is unlimited and unregulated, may include kitchen staff and managers. However, an informal arrangement by which a server "tips out" a cook or a busser at the end of a shift is unlikely to be reported to the employer and therefore unlikely to show up on our administrative data.

3 Data Sources

A key reason for the dearth of prior studies on tipping is a lack of high-quality data. We overcome this hurdle by combining several large administrative and survey datasets on both restaurants and workers, covering U.S. SU FS restaurants from 2005 through 2018. In 2017, SU restaurants accounted for about 84% of FS restaurants, 62% of FS restaurant employment, 63% of FS restaurant payroll, and 65% of FS restaurant revenue.[18]

We start with data on restaurants from the Census Bureau's Business Register (BR), Longitudinal Business Database (LBD), and Economic Census (EC). Each of these datasets includes detailed North American Industry Classification System (NAICS) codes that allow us to distinguish FS restaurants from other restaurants.[19] The EC, conducted in 2007, 2012, and 2017, includes questions directed at restaurants about service levels, which allow us to refine the assignment of industry designations.

The datasets also include other pertinent information about restaurants. First, both the EC and the BR include the business address, which we use to determine the applicable minimum wages. (We link minimum wages based on state, and in some cases city or county, as well as business size. More details on our minimum-wage data are in Appendix A.2.) Second, for SU restaurants, the BR and LBD include quarterly information on employment and payroll, and the BR also includes quarterly information on tips, as reported on IRS tax filings (Form 941). Quarterly data allow us to observe any restaurant that operated in any quarter of the year, a useful feature in an industry with a high level of business churn.[20]

[18]The SU designation is based on ownership information; therefore, an SU restaurant may be a franchise of a familiar name brand, as long as the restaurant's owner does not operate other restaurants. These statistics can be derived from data at `https://data.census.gov/table?text=7225&g=010XX00US&y=2017&n=722511:722513&tid=ECNSIZE2017.EC1700SIZESUMU`, accessed November 11, 2023.

[19]Although the NAICS code for FS restaurants changed during the period of our sample, from 722110 before 2012 to 722511 in 2012 and thereafter, the definition of the industry did not change substantively. Most non-FS restaurants are fast-food, or limited-service, restaurants, but this broad category also includes bars, coffee shops, food trucks, catering services, and more.

[20]Business Dynamics Statistics for Single-Units (BDS-SU) reports that on average over this time period 11% of SU restaurants operating in the second quarter of a given year exited by the second quarter of the following year. At the same time, 13% of SU restaurants operating in the second quarter of the year were

Finally, for most of these restaurants, the LBD includes annual revenue, also derived from IRS tax filings.

Once we identify records associated with SU FS restaurants in the BR/LBD, we use data from Form W-2 records to identify workers at those restaurants. Form W-2 provides a breakdown, by worker-job-year, of wage earnings and reported tips. These individual tax forms are filed by employers and contain individual and business tax identifiers linking workers to restaurants. Form W-2 records, however, indicate only that the worker was employed by the firm and do not assign a job to a particular restaurant location. Consequently, for workers employed by multi-unit (chain) firms, it is not possible to know which minimum-wage laws apply to which jobs, nor is it possible to know which restaurant locations within the firm report tips. Finally, in most years, we only have restaurant-level revenue for SU firms. For all these reasons, we limit our sample to SU restaurant firms.

Table 1 shows summary statistics for restaurants with and without reported tips. The first two rows show two measures of employment – employment at a point in time (we use the pay period that includes March 12) and flow employment (the number of workers who receive a Form W-2 from the restaurant). Both measure counts of individual workers, not adjusting for hours worked. The third row shows annual revenue.[21] The last two rows show payroll and tips, two size measures that may be endogenous to tip-reporting incentives. Restaurants that report tips have on average approximately 20 employees on March 12, and approximately 35 employees who work at the restaurant at some point during the calendar year. The respective numbers for restaurants with no reported tips are 6 and 11 employees. Consistent with these numbers, the average revenue for restaurants with reported tips is approximately 2.5 times the average revenue for restaurants that do not report tips, and payroll is four times larger.

entrants that had not operated in the second quarter of the prior year. See `https://www2.census.gov/ces/bds/experimental/bds2021_su_vcn4_real.csv`, accessed January 19, 2024.

[21] We use propensity-score weighting to adjust for nonrandom missing revenue. See Appendix A.1 for details.

Figure 1 shows the share of SU FS restaurants that report tips by year. The solid blue line shows the unweighted share, and the solid black line shows the job-weighted share. In 2005, two thirds of SU FS restaurants, accounting for about 87% of jobs, reported tips for at least one worker at that restaurant; by 2018, three quarters reported tips, accounting for about 91% of SU FS jobs. On average across all years, 72% of restaurants, accounting for 89.6% of all jobs, have at least some tips reported.

We distinguish between a *worker* and a *job*: the former is an individual; the latter is a worker-employer pair. Most of our analysis is at the job-year level and a worker can have multiple restaurant jobs in a year. Table 2 shows average total reported earnings for non-tip reporting and tip-reporting jobs. Restaurant jobs with reported tips have average reported earnings of nearly $9,000 annually, just over half of which is earned in tips. Jobs with no reported tipped earn slightly less, about $8,300. For context, a full-time, full-year worker who is paid an hourly rate equal to the federal minimum wage would have earned a total of about $15,000 in 2018. If she reported tips exactly equal to the federal tip credit, her reported tip earnings would be approximately $10,500 of this amount. However, many of the jobs in this sample are neither full-time nor full-year; a large share of workers in our sample have one or more additional jobs in the same year.

The solid blue line in Figure 2 shows a steady increase in the share of jobs that report at least some tips to the IRS, from about 46% to about 51%, over the period 2005–2010. After 2010, the share stabilizes at between 50% and 51%. The dashed black line shows the share of jobs with some reported tips, but restricts the sample to jobs at restaurants where at least one worker reported tips. The pattern is the same in both cases, but the dashed line is shifted up by a few percentage points, and – because the share of restaurants with reported tips increases over the sample period – the two lines converge by a couple of percentage points by 2018.

We augment the Form W-2 records with demographic information about workers from the Social Security Administration's Numerical Identification file (Numident). The Numi-

dent includes sex (male or female) and age for all individuals with Social Security Numbers (SSNs). We drop from our worker sample a small number of workers who do not match to the Numident, or whose sex or age are missing, or whose age at the time of their restaurant employment is less than 15 or more than 79. These restrictions drop workers who use Individual Taxpayer Identification Numbers (ITINs) rather than SSNs, likely due to their immigration status. This restriction applies only to our person-level analysis; employment totals at the restaurant level include all jobs.

Given evidence in other contexts that surveys may elicit more accurate reports of income because of the difference in incentives on surveys versus tax filings, we supplement the worker-job data with information from four surveys: the American Community Survey (ACS), which we are able to match to our administrative worker data at the worker-year level; the 2014 panel of the Survey of Income and Program Participation (SIPP), which we match at the worker-job-year level; the Current Population Survey Annual Social and Economic Characteristics Supplement (CPS-ASEC), which we match at the worker-year level; and the Current Population Survey Outgoing Rotation Groups (CPS-ORG), which we cannot link to the administrative data.

The ACS has surveyed about two million respondents per year since 2005.[22] Approximately 45,000 respondents per year have at least one SU FS restaurant job as identified by a Form W-2 from the same year. The ACS asks, among other things, for the worker's industry and occupation in the job in which she worked the most hours in the past week. We use the ACS linked to the workers' Form W-2 records to compute tip reporting by occupation. The occupation categories in the ACS are nuanced, distinguishing, for example, between cooks and chefs; we aggregate these detailed occupations into six occupation groups: food-preparation workers, attendants, servers, managers, drivers, and all other occupations.

The 2014 SIPP panel asked respondents to report their earnings from up to seven jobs per

[22]Sample sizes were smaller before 2005. See `https://www.census.gov/acs/www/methodology/sample-size-and-data-quality/sample-size/`, accessed May 25, 2023.

year and five income categories per job: wage and salary income, bonus payments, commissions, overtime payments, and tips. An advantage of the SIPP over the ACS is that each job is coded with both an industry and an occupation, whereas in the ACS respondents provide complete information only about their primary job. A disadvantage of the SIPP relative to the ACS is that the sample is dramatically smaller: the SIPP included approximately 36,000 respondents with jobs, most of them not restaurant workers, over the four years associated with the 2014 panel (2013–2016). We restrict our attention to approximately 900 job-years for 500 respondents who reported earnings from at least one restaurant job to the SIPP and for whom we can find a corresponding SU FS Form W-2; we then compare survey-reported tips to the corresponding value reported to the IRS.

The CPS-ORG asks all workers with job earnings whether they regularly earn "overtime, tips, or commissions." The CPS-ORG includes between 6,500 and 7,500 self-described restaurant workers per year during the entirety of our sample period, so in terms of size it is between the ACS and the SIPP.[23] The advantage of this sample is that it is significantly larger than the SIPP. The disadvantage relative to the SIPP is that the Census Bureau currently does not have permission to link the worker-level responses in the CPS-ORG to administrative data.[24] Instead, we use the CPS-ORG to create an alternative measure of tip reporting by occupation, which we can compare to the measure we create from the merged ACS/W-2 data.

Finally, for poverty calculations, we use the CPS-ASEC to obtain family income and poverty thresholds for tipped workers. We chose this survey because it provides the data used to calculate the official poverty statistics released annually by the Census Bureau. The annual supplement is administered in March and collects income for the previous calendar

[23]The Department of Labor relied on the 2018 CPS sample to determine how many workers would be affected by a rule change on the use of the tip credit; see Federal Register (2021a, Table 2, pp. 60145-60146).

[24]Other researchers have used this variable to identify tipped workers (e.g., Neumark and Yen, 2023; Clemens and Strain, 2023) but, because we cannot link the CPS-ORG to administrative data, we cannot verify that it captures the same workers as the Form W-2 data.

year. Poverty thresholds are assigned by family size and composition, also collected by the survey (Semega, Kollar, Creamer, and Mohanty, 2020). The CPS-ASEC has been linked to many administrative datasets as part of the development of the National Experimental Wellbeing Statistics (NEWS) project (Bee, Mitchell, Mittag, Rothbaum, Sanders, Schmidt, and Unrath, 2023). For our purposes, the two most important links are to Form W-2 records and to state Unemployment Insurance (UI) records from the Longitudinal Employer Household Dynamics (LEHD) database. Unlike Form W-2, UI earnings reports are quarterly, so matching the two sets of job records allows us to determine how many quarters a worker was employed at each restaurant job. Using this enhanced database, we identify SU FS restaurant workers by matching our BR/LBD restaurant list to Form W-2 of the CPS-ASEC respondents.

More information on the data is available in Appendix A.

4 Job-Level Tip Reporting

Our strategy for measuring job-level tip reporting has two parts. First, we identify tipped jobs. Second, we measure the earnings from tips. We discuss these in the subsections below.

4.1 Identifying Tipped Jobs

In this subsection, we report on a series of tests designed to identify tipped jobs. Identifying tipped jobs is critical for understanding how many, and which, jobs (and workers) are subject to specific policies targeted at these jobs, such as the tipped-subminimum wage. We show it is possible to identify the vast majority of these jobs using administrative data.

Time-Series and State-Level Variation

We begin by examining observed tip-reporting patterns for jobs over time and across state minimum-wage regimes. As noted in Section 2, tip-reporting incentives vary along both these

dimensions, but there is no evidence that the share of jobs actually receiving tips varies.

Recall from Figure 2 that the share of jobs with at least some reported tips increased from about 46% to about 51% over the period 2005–2010, and stabilized thereafter at between 50% and 51%. The increase in the share of jobs with reported tips in the early part of the time series is consistent with under-reporting of tips in those years. Although not conclusive, the stabilization in the later part of the sample is consistent with all tipped jobs reporting their tips by 2011.

Next, we compare the share of jobs with reported tips on their Form W-2 in states with a large tip credit and those in states without any tip credit. If the share of jobs with reported tips is much higher in states with a tip credit than in states without a tip credit, this could be because workers report earning tips due to pressure from employers to meet the tip credit. Conversely, if the share of jobs with reported tips is invariant to this incentive, that is consistent with accurate tip reporting by workers. We compare jobs in states without a tip credit to those in states with the federal tip credit because the federal credit is close to the maximal level. Many more states had a tipped-subminimum wage above the federal level than had a minimum wage above the federal level. Consequently, few states have a higher difference between their minimum wages (i.e., tip credit) than the federal level.

In the first column of Table 3 we compare the share of jobs with reported tips in the 14 states with continuously binding federal minimum wages to the share reporting tips in the seven states with no tip credit, where all jobs were paid at least the state minimum wage, for the period 2005–2010.[25] In states with no tip credit, 43% of jobs reported earning tips, compared to almost 47% in states with a federal tip credit. This difference is consistent with the presence of a tip credit creating reporting incentives for workers, although it could also

[25]The states in which the federal minimum wages were binding continuously from 2005–2010 are Alabama, Georgia, Indiana, Kansas, Louisiana, Mississippi, Nebraska, South Carolina, South Dakota, Tennessee, Texas, Utah, Virginia, and Wyoming. During this period, the federal minimum wage increased three times: on July 24 of 2007, 2008, and 2009. Because the federal tipped-subminimum wage did not change, the federal tip credit increased each time, starting at approximately 58.6% and rising to approximately 70.6%. The states with no tip credit are Alaska, California, Minnesota, Montana, Nevada, Oregon, and Washington.

be explained by employers preferring to staff their restaurants with more tipped occupations.

The second column shows a similar calculation for the second part of the time series, 2011–2018. The federal tip credit was 70.6% throughout this period and applied in 14 states.[26] The share of jobs with reported tips in these two types of states is almost identical: 49.3% in states without a tip credit, and 49.5% in states with the federal tip credit.

Taken together, after 2010, the flat share of jobs with reported tips and the fact that reporting at least some tips is not sensitive to the existence of a tip credit are consistent with accurate identification of tipped jobs. Across the two time periods, employers in states with a tip credit have an incentive to use tips as part of workers' compensation. This incentive is absent in states without a tip credit. Between the two time periods, reporting incentives increased due to more debit- and credit-card transactions and better reporting of these transactions.

Comparison of Tip Reporting on Form W-2 and the SIPP

Next, we compare administrative Form W-2 tip reports to tip reporting in the SIPP. If tip receipts were dramatically under-reported to the IRS primarily due to tax incentives, reporting rates should be higher on household surveys where there are no financial ramifications to reporting additional earnings.

Table 4 shows the 2×2 matrix of tip reporting to the SIPP and on the same job's W-2 filings; 70% of worker-job-year observations lie on the diagonal. Among the off-diagonal entries, only 5% of jobs reported earning tips in the SIPP but neglected to report any tips on the matched Form W-2 filings the same year. In contrast, 25% reported tips on their

[26] These states are Alabama, Georgia, Indiana, Kansas, Kentucky, Louisiana, Mississippi, North Carolina, South Carolina, Tennessee, Texas, Utah, Virginia, and Wyoming. Note that these are different from the early time period. Nebraska and South Dakota, which had followed the federal minimum wage in the early period, both increased their minimum wages in 2015 above the federal minimum wage. Kentucky and North Carolina increased their minimum wages slightly earlier than the federal government, but had converged to the federal minimum wages by 2011.

Form W-2 but not to the SIPP, supporting our view that workers who earn tips report this to the IRS.[27]

Occupation Variation

Our final check in this section analyzes tipping by occupation using several household surveys. First, we analyze tipping by occupation for 365,000 worker-year ACS respondents whose self-reported jobs were classified in the "restaurant and other food services" industry and whose records we were able to match to at least one Form W-2 issued by a SU FS restaurant, as classified by the BR, in the same year. Our prior is that, because tips are primarily given to staff who interact with customers, we should see very high tip-reporting rates on Form W-2 for occupations that interact the most with customers. In restaurants, this category of workers may include servers, bartenders, and counter staff but traditionally does not include, for example, cooks and dishwashers.[28]

Figure 3 shows the share of workers with reported Form W-2 tips for each of six ACS occupation groups. Consistent with accurate identification of tipped jobs, 85% of servers report earning tips, compared with only 19% of food-preparation workers. The imperfect alignment of tip reporting and occupation is not surprising: many workers have more than one job throughout the year (concurrently or sequentially; with a single employer or at different employers), but they can only report a single job and occupation to the ACS, so a food-preparation worker could well have had another job in which she interacted with customers and received tips. In addition, occupation reporting has known errors, and a food-preparation worker may report a server occupation even if she was never a server.

[27]The SIPP collects information about type of pay at each job using a "mark all that apply" question. This feature of the survey may contribute to a lower propensity to report tips in the SIPP relative to Form W-2 because there are no checks that prevent impatient respondents from marking only one box to move quickly to the next question.

[28]Tip sharing and voluntary tip pooling may extend to these staff, so they may still report some tips. Azar (2005) lists "restaurant waiter" as the most commonly tipped occupation, and also assigns high scores to "wine steward" and "bartender," but a lower score to "busboy." Cooks and dishwashers do not appear on his list at all.

Houseworth and Fisher (2020), in a review of the occupation-measurement-error literature, cite error rates of up to 24% in one-digit occupations and possibly above 50% in three-digit occupations. Given these error rates, the occupation groups we construct based on three-digit occupation categories are likely to falsely label a non-trivial portion of workers as servers. Thus, the fact that 85% of self-described servers also report tip income on their Form W-2 is consistent with very high tip-reporting rates.

Next, we compare the prevalence of reported "overtime, tips, and commissions" on the CPS-ORG to the prevalence of reported tips on Form W-2. Because the Census Bureau currently does not have permission to match the survey to Form W-2 data at the respondent level, this analysis uses aggregate data and includes workers not only at FS restaurants but also at limited-service (fast-food) restaurants and other eating and drinking establishments such as coffee shops; and jobs at multi-unit (chain) restaurants as well as SU restaurants. In Table 5, we compare workers who report a job in the CPS-ORG that is coded with a restaurant industry and occupation to those who report a job coded in the same way in the ACS over the same period (2005–2018). The first column shows the mean value of an indicator for whether a CPS-ORG respondent reported that she "usually receive[s] overtime, tips, or commissions"; because tips are pooled with overtime pay and commissions, this indicator may capture some workers who are not tipped. To remove errors associated with proxy reporting in the CPS-ORG, the second column restricts the sample to respondents who self-reported. Tip reporting is slightly higher for the self-reporting group. The third column uses Form W-2 reported tips for workers with ACS restaurant industry and occupation.[29] Tip-reporting rates for food-preparation workers are lower in the administrative data than

[29]The CPS-ORG asks about the worker's primary job (the one with the most hours worked last week). To maximize comparability of the ACS/W-2 data, we obtain the Form W-2 tips from the job with the highest W-2 earnings for that worker. However, differences between the CPS-ORG and ACS columns remain. Most importantly, because we do not link CPS-ORG workers to the Numident, and the CPS-ORG jobs are not restricted to those that file Form W-2, CPS-ORG figures could include informal, cash-only jobs where the distinction between "wages" and "tips" may be meaningless. This could inflate differences between survey and W-2 reports.

combined reporting rates of tip, overtime, and commission earnings in the survey, possibly due to overtime work or noise in occupation reporting. But for food attendants and servers, reported tip rates are substantially higher in the administrative data than in the survey: for example, nearly 74% of servers reported some tips to the IRS, compared to 67% who self-reported tips to the CPS-ORG. These results are consistent with administrative data correctly identifying tipped workers.

We can also check Form W-2 tip reporting by occupation using the SIPP, where respondents describe their occupations separately for each job. Our job-level match between survey and Form W-2 reports allows us to more accurately compare tip reporting by occupation. We focus on the two occupations from Figure 3 with the lowest and highest tip-reporting rates, respectively: food-prep jobs and servers. The corresponding tip-reporting rates for jobs with these occupations that match to Form W-2 records are 16% for food-prep jobs and 84% for servers. Despite the small sample, these values are extremely close to the tip-reporting rates for the same ACS occupations.

Summary of Identifying Tipped Jobs

All the exercises in this section point to the majority of jobs with tips earnings reporting earning tips, particularly in the latter part of the sample period. We do not see significant reporting differences across states based on minimum-wage laws, nor do we see continued increases over time after 2010, consistent with near-complete reporting. At the occupation level, the vast majority – 85% – of servers report earning tips on a restaurant Form W-2. Finally, we do not observe significant under-reporting of tips on Form W-2 relative to household surveys; if anything, the Form W-2 data are more likely to have reported tips than the surveys.

4.2 Measuring Earnings from Tips

We now turn to measuring earnings from tips. Although we are able to identify most tipped jobs in administrative data, we believe that the dollar value of tip earnings is misreported.

Mass Point

The clearest signal of possible misreporting of tips is a mass point, or bunching, of reported tips at thresholds that are mandated by law. We start by examining tips as a share of total earnings for jobs in the 14 states that used the federal minimum wages continuously from 2011–2018. Define the tip-credit ratio as $\frac{\text{tip credit}}{\text{minimum wage}}$. Given the federal minimum wage of \$7.25/hour, tipped-subminimum wage of \$2.13/hour, and therefore a tip credit of \$5.12/hour, the federal tip-credit ratio during this period is approximately 0.706. If a worker in a state where the federal tip credit applies reports tips equal to the tip credit for every hour she works, her ratio of tips to total earnings will equal 0.706.

Figure 4 shows the distribution of jobs' share of earnings accounted for by reported tips $\left(\text{i.e., } \frac{\text{tips}}{\text{wages+tips}}\right)$ in the years 2011–2018 in the 14 states with the federal tip credit, with a vertical line at the federal ratio. A large mass of jobs report tips exactly in proportion to the tip credit, and relatively few jobs report earning less in tips.

To expand our analysis to include additional years as well as jobs in states with different tip-credit levels, we normalize the tip share of earnings by the tip-credit ratio:

$$\frac{\left(\frac{\text{tips}}{\text{wages+tips}}\right)}{\left(\frac{\text{tip credit}}{\text{minimum wage}}\right)}$$

This normalization allows us to pool jobs in states subject to the federal tip credit with jobs in other states with a tip credit; restaurants in approximately half of all U.S. states are subject to tip-credit ratios that are lower than the federal ratio due to higher state subminimum wages. A worker who reports tips equivalent exactly to the tip credit in her state in every hour she works in a calendar year will have a normalized tips-to-earnings ratio

equal to one. Jobs with fewer tips reported than the tip-credit ratio requires will have a normalized tips-to-earnings ratio less than one, and those with reported tips in excess of the tip-credit ratio will have a normalized tips-to-earnings ratio above one.[30]

Figure 5 shows the distribution of the normalized tips-to-earnings ratio for jobs with reported tips in states with a tip credit, using data for 2005–2018. As is evident in the figure, a disproportionate share of jobs report exactly enough tips to cover the tip credit: 12% of observations lie between 0.95 and 1.05. This pattern is reminiscent of mass points of self-employment income at thresholds that are consistent with either misreporting or very careful tax planning. It also echoes bunching of reported income of workers in Hungary at the minimum wage, when, in fact, a substantial share earn at least twice that amount, as documented by Bíró, Prinz, and Sándor (2022). The bunching of the normalized tips-to-earnings ratio at 1 is consistent with many workers employed in these jobs misreporting their tips to their employer and the IRS.

Household Surveys

We supplement the bunching evidence with a comparison of survey tip earnings with reporting on Form W-2. Bollinger, Hirsch, Hokayem, and Ziliak (2018, Table 6) reports earnings differentials, by occupation, between survey responses on the CPS-ASEC and tax filings.[31] Unlike the CPS-ORG, the CPS-ASEC does not ask whether a worker earns tip income, but the occupation group with the fourth-highest discrepancy is "food prep and serving related workers," which includes tipped servers, suggesting that tips may be reported differently on surveys and tax forms.

Because the SIPP explicitly collects tip earnings, we can compare the dollar amount tipped workers report to the SIPP with the amount on their Form W-2. We focus on the 29% of SIPP restaurant worker-job-year observations with positive reported tips on the SIPP

[30]The ratio is not defined for jobs in states without a tip credit.

[31]The paper was published as Bollinger, Hirsch, Hokayem, and Ziliak (2019).

as well as on Form W-2. Among these approximately 250 worker-job-years, 65% reported more tip income to the SIPP than they reported on Form W-2.

Summary of Measuring Earnings from Tips

The bunching of reported tips at the tip credit suggests that tip income is misreported, but, on its own, does not allow us to determine whether tips are under- or over-reported. Misreporting could be due to strategic considerations – specifically, a desire to avoid paying income and FICA taxes on excess tips – but it could also be for more innocuous reasons, such as not tracking cash tips carefully and "guesstimating" them at the end of a pay period or quarter. Tip pooling may also push workers towards reporting exactly enough tips to make up the tip credit: workers cannot be required to share tips if they earn less than the tip credit, and those who earn more tips may end up subsidizing those who earn less. Wage theft or "tip skimming" (employers illegally taking tips from workers) could also cause the observed bunching.[32]

The evidence from the SIPP suggests tips may be under-reported to the IRS, but the sample is too small to be conclusive.

We return to this issue in Section 5.2, after we discuss restaurant-level measures of tips. Because we have restaurant-level revenue, calculating restaurant-level reported tips relative to revenue allows us to benchmark reported tips against tipping norms.

5 Restaurant-Level Tip Reporting

Having exhausted our job-level data sources, we turn to measures at the restaurant level. We start with measuring the presence of tipped workers, followed by measuring the value of tips at the restaurant level.

[32]Surveys designed specifically to assess the prevalence of wage theft have found high reported rates among restaurant workers; see Dickson and Wood (2023) and Bernhardt, Milkman, Theodore, Heckathorn, Auer, DeFilippis, González, Narro, Perelshteyn, Polson, and Spiller (2009).

5.1 Measuring the Presence of Tipped Workers

Fundamental to our understanding of FS restaurants in the U.S. is the assumption that tipping occurs at all of them. If an employee seats patrons, offers them menus, takes their orders, and brings the food to the table, a tip is generally expected; even if not all patrons leave tips, enough do that, at year's end, collectively, workers at the restaurant will have received some tips.[33] If industry codes are reliable – that is, we have correctly identified FS restaurants – and if all workers who earn tips report at least some of them, we should see very few restaurants with no tips in our data. Despite this, approximately 28% of SU FS restaurant-years in our data have no reported tips. In this section, we attempt to determine what share of these non-reporters are not FS restaurants and what share simply fail to report tips.

We take a two-pronged approach to decomposing the non-tip reporting establishments into misclassified establishments and non-reporters. First, we conduct a hand audit of 1,000 randomly-selected restaurant-year observations to assess the error rate in NAICS coding. We restrict our audit to observations in the three EC years covered by our sample (2007, 2012, and 2017) and utilize information about the name of the business and whether the business was surveyed by and responded to the EC. Very small restaurants are often either not surveyed by the EC or do not respond, both of which increase the likelihood that their BR/LBD industry codes are inaccurate. We include a mix of restaurants that reported tips in some but not all years and restaurants that never reported tips. We conclude that approximately 75% of the establishments in our audit are plausibly FS restaurants.[34]

The second approach we take is checking whether the share of non-tip-reporting restaurants is sensitive to reporting incentives. We examine three incentive margins: minimum-

[33]There are some FS restaurants that do not allow tipping, but casual observation suggests that they are exceedingly rare.

[34]This is a subjective and inexact exercise. In many cases, it is impossible to discern from the business name whether a business is a restaurant at all, let alone a FS restaurant: "Basker, Foster, and Stinson" could be almost any business.

wage regime, employer size, and year. If an establishment is not a FS restaurant and its workers do not earn tips, it should not be sensitive to these incentives. If, conversely, increased reporting incentives increase the share of establishments that report tips, we conclude that these are likely FS restaurants.

The minimum-wage regime affects reporting incentives through the channel examined in Section 4, namely that a tip credit provides an incentive for employers to ensure that their workers report tips. We examine whether a higher share of restaurants have employees reporting tips in states with a tip credit.

Employer size affects reporting incentives because the IRS revenue-share rule only applies at restaurants with at least ten employees on a typical business day, and because, after 2011, Form 1099-K reporting is based on the value and number of card transactions. We have shown in Table 1 that restaurants that do not report tips are significantly smaller than restaurants that do report tips: they account for more than a quarter of restaurant-years, but only a tenth of job-year observations. We do not know how many workers are on duty on a typical business day at the restaurants in our sample, nor the value of credit-card transactions, so we use a cutoff of 20 employees to split the sample of restaurants into small and large employers. Small employers with fewer than 20 employees are not affected by the revenue-share rule and hence have a lower incentive to ensure tip reporting by their employees. In addition to the IRS revenue-share rule, smaller restaurants may be less likely to report tips for various other reasons. They may be more informally managed and lack the expertise to comply, or they are not subject to FLSA and minimum wage laws, or they may assume they are less likely to be audited. Among large employers affected by this rule, the incentive may be more salient for restaurants operating in a state without a tip credit.

Finally, time plays a role for several reasons. As more transactions move to payment cards and electronic apps, we expect increased tip reporting both because card payments are easier to track by employers and workers, and hence may be better reported regardless of incentives, and because the third-party credit- and debit-card reporting instituted by Housing

and Economic Recovery Act increased restaurants' incentives to report tips accurately to the IRS after 2011. In addition, the FLSA may apply to more restaurants over time, as restaurants' nominal revenue has increased with inflation.

Table 6 shows the share of restaurants reporting tips in twelve different bins. The first two columns show reporting shares by restaurants in the seven states without a tip credit, first for the early part of the sample (2005–2010) and then for the later part (2011–2018). In each column, we show the overall reporting share, as well as the share reporting tips among restaurants with fewer than 20 employees and the share reporting with 20 or more employees. The last two columns provide the same information for states with the federal tip credit. Note that the set of states differs slightly between the early and late parts of the sample (as does the federal tip credit).

In the early time period, only 60% of restaurants in the seven states with no tip credit, where reporting incentives are lower, reported tips. This compares with 68% of restaurants in the 14 states with the federal tip credit, where the large tip credit creates a strong reporting incentive for restaurants. In both sets of states, restaurants with 20 or more employees are very likely to report tips, and small restaurants are much less likely to report tips. The difference between restaurants in the two sets of states are attenuated for large restaurants.

Reporting rates increase for all categories in the later period: 68% of restaurants in states with no tip credit report tips, as do 74% of restaurants in states with the federal tip credit. The differences between small and large restaurants shrink slightly, as do differences across tip-credit regimes, but the general pattern remains: restaurants with more incentives to report tips are more likely to report tips.

The incentives are strongest for large restaurants in states with a federal tip credit in the second part of the sample period: here, all three incentives – the tip credit, the IRS rule, and increased reporting transparency – all operate in the same direction. The lower-right cell in Table 6 shows that the reporting rate for these businesses is 91%.

We use the following thought experiment to determine how much of non-reporting of tips

is due to misclassification of restaurants, rather than non-reporting by true FS restaurants. First, recall that 28% of restaurant-year records have no reported tips, and that our hand audit suggests that about 25% of them – or about 7% of all restaurant-year records – are potentially misclassified. Second, from Table 6, of the most-incentivized restaurants, under 9% fail to report any tips. If we assume that the vast majority of businesses that do not report tips despite the triple incentive to do so do not collect tips – that is, they are not FS restaurants – and, further, that the rate of misclassification at large restaurants in states with a federal tip credit in the latter part of our sample is representative of all restaurants in all years of our sample, then our two approaches provide very similar results. In Section 6 we use the estimate that 7% of observations in our restaurant-year panel belong to other types of businesses – coffee shops, bars, barber shops, insurance companies, etc. The remaining 21% of observations with no restaurant-level tips belong to FS restaurants, and these are sensitive to reporting incentives.

Reconciling Job- and Restaurant-Level Results

The evidence in Section 4.1 seems consistent with accurate identification of tip-earning *jobs*, particularly after 2011. We base our conclusion on a series of checks, including the invariance of tip reporting to reporting incentives both over time and across state minimum-wage regimes; consistency of reporting with our priors on the occupational distribution of tipped workers; and a favorable comparison of tip reporting on workers' Form W-2 with reports of tip earnings in household surveys.

In this section, however, we show that many smaller FS restaurants have no tip reporters, and that this margin is sensitive to reporting incentives such as the presence of a tip credit for workers. These non-tip-reporting businesses account for 10.5% of worker-job-years. Applying our estimate that three quarters of these businesses are indeed FS restaurants, we calculate that about 8% of all worker-job-year observations are employed at FS restaurants that do not report tips. Assuming that half of these jobs are tipped implies that about 4% of all

SU FS workers receive tips but fail to report any of them. These jobs are concentrated in small restaurants that are disproportionately located in states with no tip credit, and their numbers have been declining over the period of this study.

Another conclusion from considering both worker and restaurant reporting rates is that the level of tip reporting varies by person, or job, whether the worker reports *any* tips depends more on the characteristics of the employing restaurant, namely its geographic location and size.

5.2 Measuring Restaurant-Level Tips

Tips Relative to Revenue

Before we turn to the data at the restaurant level, how much should we expect in tips?

Azar (2020) cites various sources that *recommend* a tipping range of 15–25% of the check amount. Public data we obtain from two consumer surveys and a payment application are on the lower end of this distribution, much closer to 15% than 25%.

The first consumer survey, FoodAPS, is described in Appendix A.2; it references nearly 2,000 meals, almost all of which took place in 2012; the average tip rate in that survey is 14.6%. In another consumer survey from late 2019, shortly after the end of our sample period (but, importantly, before the beginning of the Covid-19 pandemic), 20% of respondents agreed with the statement "I recently left no tip at a restaurant with table service" and only 82% said they regularly tip waitstaff at restaurants (TD Ameritrade, 2020).[35]

The payment data from the Square payment app are also described in Appendix A.2; the exact date of transactions is not given, but the data were published in 2014 and likely refer to transactions in 2013–2014, mostly at small restaurants. In the Square data, only about 57% of transactions include tips; among transactions that were tipped in the application, the

[35]The survey included approximately 1,000 U.S. adults ages 23 and older who had assets of at least $10,000. Consistent with our other data sources, among those who did leave a tip, the median tip rate is in the 19-25% bin, with 43% tipping 18% or less and only 11% tipping 26% or more.

average tip rate was approximately 16.5%. We cannot distinguish between cases in which the tip was paid in cash, cases in which the transaction was not at a FS restaurant, and cases in which the patron left no tip at a FS restaurant.

With this context in mind, Figure 6 shows the time series of the mean ratio of tips to revenue.[36] This mean increased from about 7% in 2005 to about 9% by 2018, and shows little evidence of stabilizing. The ratio could be increasing because the rate at which consumers tip is increasing, but it could also be increasing because reporting rates are higher holding actual tips at a fixed level.

Table 7 shows means of tips/revenue for various subsamples as in Table 6. The means range from 6.5% in the early part of the sample at restaurants not subject to the tip credit to 11% in the later part of the sample at large restaurants subject to the tip credit. However, across the board, these means fall far short of 15%, the lowest bound from the external sources cited above.

Mass Point

Because larger FS restaurants are required by the IRS revenue-share rule to report tips equal to at least 8% of revenues, we can examine the distribution of the ratio of tips to revenue at the restaurant level to check for a mass point around 8%. *Ex ante*, the 8% threshold may be less salient than the FLSA tip-credit threshold because of the way tips are reported: reporting at the restaurant level is an aggregation of reporting by workers. Whereas the FLSA reporting threshold is relatively easy for workers to observe and meet – it is specified on an hourly basis, and they can tally up their hours to determine the reporting level required by shift or week – it is less obvious how their reporting could be tailored to meet an aggregate threshold that also depends on the restaurant's revenue, which they do not observe. Nevertheless, there is some evidence that this threshold does matter.

[36] We use weighting in this figure to adjust for the fact that that revenue is missing for some observations in the sample. See Appendix A.1 for details.

The solid blue line in Figure 7 shows the distribution of the (tips/revenue) ratio using data from 2005–2010, and shows a small spike at 8%. The dashed black line in Figure 7 shows the distribution of (tips/revenue) after the implementation of the Housing and Economic Recovery Act in 2011. Consistent with the Act's increased reporting incentives, the distribution shifts to the right and the mass point at 8% diminishes, implying that reported tips grew faster than restaurant revenue between the two time periods.

Because more tips are reported in the later time period and the mass point is diminished, we examine this period in more detail. Specifically, we recreate the distribution of the ratio (tips/revenue) for four smaller samples: small vs. large restaurants, and restaurants subject to the federal tip credit vs. those not subject to any tip credit. The distributions are shown in Figure 8. As in Section 5.1, we use a cutoff of 20 employees over the course of a year as a rough proxy for restaurants having at least ten employees on a typical business day (and therefore subject to the IRS revenue-share rule).

The left panel shows restaurants with fewer than 20 employees. These restaurants likely do not have the minimum employment for the IRS revenue-share rule to bind and are also less likely to receive Form 1099-Ks, so we do not expect to see a mass point at 8%, and in fact the vast majority of the mass is well below the 8% line, consistent with Table 7. For restaurants located in states without a tip credit, represented by the solid blue line, the incentive to report tips is particularly low, and the mass point is shifted even lower than for restaurants subject to the federal tip credit, represented by the dashed black line.

The right panel shows restaurants with 20 or more employees, for whom the revenue-share rule is much more likely to apply. However, it may not bind: restaurants subject to the high federal tip credit may already be reporting sufficient tips to comply with the revenue-share rule because of FLSA incentives. Indeed, the dashed black line shows there is virtually no spike at 8% for these restaurants. In contrast, the IRS revenue-share rule is likely to be much more salient for large restaurants not subject to a tip credit, and the solid blue line shows that for this subset of restaurants, there is still a mass point around 8% even

in the second part of the sample period.

Reconciling Job- and Restaurant-Level Results

In Section 4.2, we noted that we cannot definitively say whether individual workers under-report or over-report their tips. However, our analysis at the employer level implies that tips are under-reported, given that the ratio of tips to revenue falls short of U.S. tipping norms, even at restaurants where some workers report earning tips.

5.3 Restaurant Size and Job-Level Tip Reporting

If the level of tip reporting at the restaurant level depends, in part, on the applicable tip-credit regime, it stands to reason that the salience of the revenue-share IRS threshold could also impact job-level tip reporting. To test this idea, we create two distributions of job-level normalized income shares from tips, similar to Figure 5: one for jobs at employers with fewer than 20 employees, and one for jobs at employers with 20 or more employees. These are shown in Figure 9.

Both distributions show bunching at the tip-reporting threshold, but this bunching is stronger for jobs with reported tips at small employers, shown as a solid blue line, where 14.4% of observations are between 0.95 and 1.05, compared with 11.7% of worker-job-year observations at large employers, shown as a dashed black line. The distribution of normalized tips to earnings is shifted to the right for jobs at large restaurants, indicating that workers report more tips relative to their wages when they work for larger employers. Although it is possible customers tip more generously at large restaurants, it is also possible that restaurants that are required to meet the IRS revenue-share rule monitor tip reporting more closely and demand a higher rate of tip reporting by their workers.

6 Impacts of Tip Under-Reporting

As the title of our paper suggests, one of the contributions of this paper is establishing an updated measure of the size of the tip of the iceberg: the share of tips that is observed in tax records. The IRS has estimated that the tip of the iceberg is 45% (Internal Revenue Service; Research, Applied Analytics and Statistics, 2022, pp. 2, 18), but we find that it is about 60%. We then explore some of the implications of tip under-reporting for the measurement of poverty, which relies on precisely measured compensation.

6.1 Industry-Level Missing Tips

In this section, we perform two back-of-the-envelope quantifications of missing tips in the FS restaurant industry. First, we calculate the aggregate value of missing tips and estimate the shares missing due to restaurant non-reporting vs. under-reporting. Second, we calculate under-reporting per job. These exercises are imprecise due to data limitations, but we believe they are nevertheless informative about the potential magnitude of total unreported tips.

Allocating Missing Tips to Restaurants

To calculate what share of tips is missing from tax reports, we need a prior on actual tips that is independent of tax filings.

First, we need an estimate of the number of SU FS restaurants with no reported tips. Table 1 shows that our sample includes on average about 51,000 restaurants with zero reported tips per year. Based on our analysis in Section 5.1, we assume that 75% of these non-reporters are FS restaurants, or about 38,000 restaurants per year. We further assume that the means in column 2 of Table 1 are representative of this sample.

Second, based on the discussion in Section 5.2, we use 15% as our benchmark for the true tipping rate. This rate can be rationalized by an average tip rate on *tipped transactions*

of 18.75% and a stiffing rate of 20%, or an average tip rate of 16.67% and a stiffing rate of 10%.

We use this tip rate to attribute missing tips to restaurants with and without any reported tips. Table 1 shows that our non-reporters have an average annual revenue of about $390,000, for a total of $14.5 billion in revenue. We assume that 95% of revenues, or $13.8 billion, were subject to tips in an average year.[37] If actual tips equal 15% of this amount, restaurants with no reported tips are missing $2 billion in tips per year.

For restaurants that do report some tips, the same table shows average annual revenue is about $950,000, or almost $125 billion in total annual revenue across 131,000 tip-reporting restaurants. Applying the same calculations as above – assuming 95% of revenue derives from transactions normally subject to tips and a 15% average tip rate on those transactions – we would expect $17.8 billion in total tips, but see only $11.8 billion reported, a shortfall of approximately $6 billion per year.

All told, FS restaurants reported tips on Form W-2 amounting to $24.2 billion per year on average over the period 2005–2018. Our sample of SU restaurants contributed $11.8 billion to this total. (The remaining $12.4 billion is due to multi-unit restaurants.) Our calculations based on restaurant revenues imply that SU FS restaurants failed to report approximately $8 billion in tips. We attribute about a quarter of the missing tips ($2 billion) to restaurants that report no tips, and three quarters ($6 billion) to restaurants that under-report their tips. If our assumptions are correct, workers at these SU FS restaurants collectively reported 60% of all tips earned. This is quite a bit higher than the IRS's estimate that only about 45% of tips are reported (Internal Revenue Service; Research, Applied Analytics and Statistics, 2022).

[37]Published EC tables for 2007 and 2012 show that approximately 5% of FS restaurant revenue is from food and beverage sales for consumption off the premises (take-out orders), and 1–3% is from miscellaneous goods and services, including non-food merchandise and catering. See `https://www2.census.gov/programs-surveys/economic-census/data/2007/sector72/` and `https://www2.census.gov/programs-surveys/economic-census/data/2012/sector72/`; last accessed January 30, 2024.

Missing Tips per Job

To estimate the amount of under-reported tips per job, we start by calculating a ballpark estimate of the number of jobs missing tips. From Table 1, SU FS restaurants that report no tips account for about 540,000 jobs. If we assume that 25% of these jobs are at establishments that are not actually FS restaurants, we are left with about 400,000 employees. If non-tip-reporting restaurants are similar to reporting restaurants, then approximately 55% of their workers actually earn tips, for a total of 220,000 jobs per year where the compensation includes tips but this income is not reported. We also see in Table 1 that tip-reporting restaurants account for 4.6 million jobs, of which 2.5 million report some (but not all of their) tips.

We combine these ballpark estimates of the numbers of jobs affected by each type of mis-reporting with our estimates of the total amount of missing tips due to reporting shortfalls. This allows us to calculate average under-reporting of tips per job. If 220,000 jobs per year at restaurants with zero reported tips actually collectively earn $2 billion in unreported tips, then, on average, these non-reporting jobs earn about $9,400 in tips per year. In restaurants with some tips reported, we estimate that, all together, workers under-report tips by $6 billion per year, or an average of about $2,350 per worker-job per year. This under-reporting amounts to a third of tips received by workers who report some tips.

In Table 2, we showed that reported tips accounted for approximately $4,700 of total earnings at the average tip-reporting job, and total earnings per tipped job were just shy of $9,000, a few hundred dollars more than average job-level earnings for non-tipped jobs. Adding unreported tips increases mean income for tipped jobs to about $11,400.

Multi-Unit Restaurants

The analysis in this paper focuses on SU FS restaurants, for which we have higher-quality and higher-frequency data than for multi-unit restaurants. Among other advantages, we have annual revenue for SU restaurants, and we are able to match workers to specific work

locations. However, a complete analysis of missing tips cannot ignore multi-unit restaurants. We use some of our results, combined with public data on multi-unit FS restaurant revenues in EC years, to supplement the missing-tips calculations.

On average in 2007, 2012, and 2017, aggregate multi-unit FS restaurant revenue was about $99.4 billion in real 2018 dollars.[38] If we assume that these three years are representative of the full sample period and apply the same assumptions as above – namely, that 95% of this revenue is subject to tips and that tips amounted to 15% of the revenue subject to tips – then workers at multi-unit FS restaurants should have received about $14.2 billion a year in tips.

From our calculations with confidential data, the dollar amount of tips reported by multi-unit FS restaurants to the IRS averaged about $12.4 billion per year over the period 2005–2018, implying that about $1.8 billion in tips are missing from multi-unit restaurants on average each year. If correct, this calculation implies that workers at multi-unit FS restaurants collectively reported more than 85% of their tips, compared with SU restaurants, where only 60% of tips are reported.

Why might reported tip rates be higher at multi-unit restaurants? We can only speculate on this point. One possibility is that industry-misclassification rates are lower for multi-unit firms, i.e., a larger share of the multi-unit establishments identified as FS restaurants are, in fact, FS restaurants, relative to the share of SU restaurants. And conditional on being true FS establishments, multi-units might be more likely to have at least some tips reported given that they are often larger in size. It could also be that tipping is more common or more generous at multi-unit FS restaurants. Another possibility is that, holding tipping constant, tip reporting is higher at chains. This could be because larger enterprises are operated more formally, with better record keeping (Bloom, Lemos, Sadun, Scur, and Van Reenen, 2014);

[38] Data from `https://www2.census.gov/programs-surveys/economic-census/data/2007/sector72/EC0772SSSZ3.zip`, `https://data.census.gov/table/ECNSIZE2012.EC1272SSSZ3?text=7225&g=010XX00US&y=2012&n=722511`, and `https://data.census.gov/table/ECNSIZE2017.EC1700SIZESUMU?text=7225&g=010XX00US&y=2017&n=722511`, last accessed March 21, 2024.

because a larger share of workers are subject to FLSA minimum wages; or because payments are more likely to be made using credit cards.

Caveats

The above estimates of tip under-reporting should be taken with a grain of salt. (Waiter! More salt, please.) Most importantly, we do not mean to imply that every worker under-reports tips. Quite the contrary: we show that there is significant heterogeneity, both in who reports and in how much tip earnings they report, across different state minimum-wage regimes and restaurant size categories, and over time. Instead, the averages should be interpreted as such, meaning that for each worker who fully reports tips or even over-reports tips, some other worker is likely under-reporting even more than the average.

In addition, the calculations rely heavily on several assumptions. First, we assume that 75% of non-tip reporting establishments in our sample are FS restaurants. If only half of them are FS restaurants, then these restaurants' missing tips decrease from $2 to $1.4 billion. Some establishments with positive, but low, reported tips may also be mis-identified as FS restaurants, in which case our estimate of under-reported tips could also be inflated. Second, we assume a net tipping rate of 15% at both tip-reporting and non-reporting restaurants. A 10% net tipping rate would produce a shortfall of only $1.4 billion, only $16 million of which would be missing from tip-reporting restaurants. Finally, it is possible that the actual tip rate, and the share of jobs that earn tips, is different at restaurants with and without reported tips, in which case our attribution of missing tips to the two groups of restaurants and their workers is inaccurate.

2018 Missing Tips

Although we have not calculated missing tips on a year-by-year basis, we do compute them separately for the final year of our sample: 2018. The latter part of our time period was one of growth for the restaurant industry, with more SU FS restaurants operating, and

average employment and revenue per restaurant rising; see Table 8. This growth caused industry-level revenue to increase, leading us to expect an increase in aggregate tips.

In keeping with our earlier findings, we estimate that reported tips as a share of all tips increased in 2018 relative to the time series average, meaning that reported tips grew faster than unreported tips. This is because the share of restaurants not reporting any tips fell and at restaurants that already reported some tips, workers reported more tips relative to revenue. We estimate that in 2018, 64% of all tips earned were reported, up from 60% on average over the full time series. Using the same assumptions as above, we find that, on average, workers at SU FS jobs earned about the same dollar value of tips in 2018 as the time series average – $7,100 per year in real 2018 dollars – but reported approximately $400, or 8%, more to the IRS.

6.2 Impact on Poverty Measurement

Is under-reporting of tip income significant enough to affect poverty measures? If tipped workers live in families with incomes just below the poverty line, and if under-reporting of tips is a feature of household surveys as well as administrative data, then it is possible that the measured poverty rate of tipped workers is inflated. To investigate this possibility, we use the 2019 CPS-ASEC, the official survey instrument for measuring poverty in the United States, combined with Form W-2s and UI earnings reports in the recently developed NEWS database. We identify tipped restaurant workers in the survey as those who have at least one Form W-2 with reported tips from a SU FS restaurant. The NEWS database has a reference year of 2018, so we focus on that year.

We begin by calculating a poverty rate for tipped SU FS workers using the official poverty methodology, which relies on the standard CPS-ASEC measure of income. We then vary the source and amount of job earnings for SU FS restaurant workers to see how poverty rates for tipped workers are impacted. Finally, we re-calculate the national poverty rate to see how overall poverty would be impacted by changing income measurement for restaurant

workers. The first set of statistics show how poverty analysis for a particular group could change; the second set show how changes for this group would impact the headline poverty rate for the country.

The official poverty rate calculation compares total family income as reported to (or imputed by) the CPS-ASEC to the poverty threshold defined for each family based on family size and composition. Families that fall below this threshold are considered poor. Rates are the weighted mean of the poverty indicator for the CPS-ASEC sample. The CPS-ASEC publishes poverty rates both for all *people* and all *workers*. The poverty rate for all workers is 5.1% (Semega, Kollar, Creamer, and Mohanty, 2020, Figure 8). Replicating this calculation for tipped SU FS gives a poverty rate of 11.7% for these workers in 2018, more than double the poverty rate for all workers.

We supplement this baseline calculation with two alternative measures. For the first version, we modify family income by replacing restaurant workers' earnings from the CPS-ASEC survey with their earnings as reported on Form W-2. Using this alternative, administrative, income measure allows us to create what we call an "administrative" poverty measure. In the second, we add our estimate of missing tips per quarter of work to each tipped job-quarter, and create an "enhanced administrative" measure of family income and poverty.

For the "administrative" measure, we replace total CPS-ASEC earnings with total 2018 Form W-2 earnings for respondents who were SU FS restaurant workers and were classified by the survey as working for an employer. (We do not replace earnings for family members not employed at a SU FS restaurant, nor alter income measures from other sources, such as self employment.) This substitution allows us to estimate what the poverty rate would have been if SU FS restaurant workers reported the exact same wage and tip earnings to the CPS and the IRS. Substituting Form W-2 earnings for CPS-ASEC earnings can increase or decrease family income, depending on whether these workers' combined Form W-2 earnings are higher or lower than their survey earnings, and can therefore either increase or decrease the measured poverty rate for restaurant workers. In practice, we find that using

this "administrative" measure raises the poverty rate for tipped SU FS restaurant workers from 11.7% to 13%, but this increase is not statistically significant.

The "enhanced administrative" measure adds our estimate of tip under-reporting in 2018 from Section 6.1 to the "administrative" family income above. Our calculations for 2018 show that, on average, annual tips for SU FS restaurant workers were under-reported by approximately $2,000, or about $800 per quarter.[39] For each tipped worker in the CPS-ASEC, we multiply this estimate of missing tips per job-quarter by the actual number of job-quarters the tipped worker was employed at any SU FS restaurant in 2018, which we obtain from UI records. By construction, adding tips can only reduce poverty because our adjustment only increases family income. However, the degree by which the poverty rate falls depends on how close the tipped-worker families are to the poverty threshold. This "enhanced administrative" method yields an estimated poverty rate of 10.4%, which represents a statistically significant decrease of 2.6 percentage points from our "administrative" measure and a statistically insignificant decrease of 1.3 percentage points – or 11% – from the standard CPS measure for tipped restaurant workers.

Our goal in this section is not to provide a definitive alternative measure of poverty, but to demonstrate that tip under-reporting has the potential to affect poverty measurement. As in Section 6.1, this estimate ignores possible heterogeneity. We attribute the same dollar value of missing tips to each job-quarter, independent of the employer's revenue and total reported tips, and also independent of the worker's and employer's reporting incentives. In practice, some workers' earnings are likely reported much more accurately than others. If workers' distance from the poverty line is correlated with the dollar value of their unreported tips, possibly because of patterns in the location or type of restaurants that employ them, then the enhanced administrative poverty measure may have systematic errors.

Although the impact of missing tips on the measured poverty rate of tipped workers is

[39]The average SU FS restaurant job in 2018 lasted for 2.5 quarters.

potentially significant, the effect of missing tips on the U.S. population poverty rate is likely to be small because tipped restaurant workers represent only a small share of all people close to the poverty line. The official poverty rate for all people in 2018 was 11.8% (Semega, Kollar, Creamer, and Mohanty, 2020, Figure 8). Because of the small weight of tipped SU FS workers in the total population, using either of our two alternative measures of family income in place of standard CPS-ASEC income, we only decrease the overall poverty rate in 2018 by approximately 0.1 percentage points.[40]

7 Concluding Remarks

Accurate reporting of income is essential for national statistics, including inequality and poverty measurement. Researchers, policymakers, and auditors have been concerned that tips, in particular, are often under-reported. For example, the IRS is concerned about both non-reporting and under-reporting of tips by restaurants.[41] The IRS assumes that workers report only 45% of their tip income (Internal Revenue Service; Research, Applied Analytics and Statistics, 2022, pp. 2, 18). This may be an improvement over historical reporting rates: Robertson, Quinn, and Carr (2006) cite a 1998 IRS estimate that fewer than 40% of tips are reported.

Our analysis suggests a slightly more nuanced picture. We find evidence consistent with a very high propensity to report tips by tipped workers, and our estimates of missing tips are slightly more positive than IRS assumptions. A back-of-the-envelope calculation suggests that on average about 60% of all tips received by workers at SU FS restaurants between 2005 and 2018 were reported on tax forms. We base this calculation on a combination of job- and

[40]The difference between the official poverty measure and our "administrative" measure is not statistically significant. The difference between our two alternative measures is very small – less than 0.1 percentage points – but it is statistically significant.

[41]Dimsdale (2010) cites an IRS estimate that only 25% of food and beverage businesses that should be filing tip-disclosure forms do so. Adelsheim and Zanetti (2006) report on an intervention that aimed to increase tip-reporting compliance of restaurants in Chicago in 2003.

employer-level data, and on several assumptions, the most consequential of which is that actual tips average approximately 15% of all sit-down meal revenue at these restaurants. Tip-reporting rates appear to be even higher at multi-unit FS restaurants.

In industries, such as restaurants, where tips are prevalent, accurate earnings measurement is inherently difficult. Because most sources of earnings data – both administrative sources and surveys – combine tip and wage income, total earnings may often be underreported for tipped workers. As restaurant workers disproportionately fall at the lower end of the earnings distribution, and tipped restaurant workers are more than twice as likely to be in poverty than other workers, this shortfall may impact poverty measurement. We demonstrate that adding tips to restaurant workers' measured income can impact their estimated poverty rate.

These findings have implications for a longstanding debate on the relative merits of administrative income data and survey data. Early comparisons of different earnings sources defined measurement error in surveys as the difference between the survey and tax reports (see, e.g., Mellow and Sider, 1983; Bound and Krueger, 1991; Gottschalk and Huynh, 2010). Later work, including Kapteyn and Ypma (2007) and Abowd and Stinson (2013), called into question the assumption that administrative data were always more reliable than survey data and built statistical models that allowed for errors in both sources. Current research on poverty measurement by Bee, Mitchell, Mittag, Rothbaum, Sanders, Schmidt, and Unrath (2023) and Meyer and Wu (2024) has fully embraced the idea that no source of earnings data is entirely correct. Instead, these papers focus on how to choose the more accurate source of earnings for different types of workers. Likewise, Chenevert, Klee, and Wilkin (2016) discuss whether replacing survey earnings reports with administrative data could help mitigate non-response bias and improve earnings imputation.

Our research offers a cautionary note for the debate over data sources. In surveys where we can observe something about tips – CPS-ORG and SIPP – there does not appear to be a clear advantage to either survey or administrative data. Tip-income mismeasurement cannot

be solved simply by using an alternative job-level data source. Absent a measure of earnings that more accurately and consistently captures tips, researchers should be aware that if surveys follow the same trend as tax data and more tips are reported over time, measured poverty rates among restaurant workers may be impacted for purely reporting reasons.

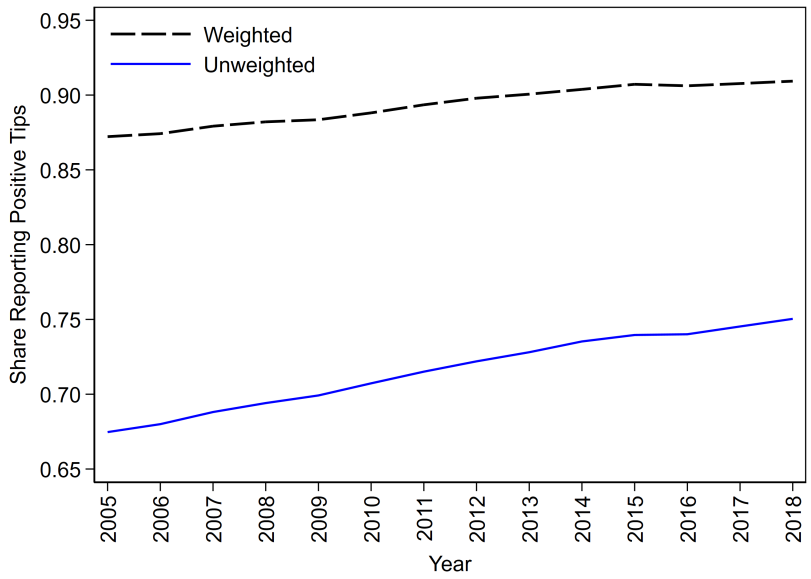
Future survey development efforts could consider whether there are design decisions that would improve the collection of tip income for workers who report restaurant employment. Higher-quality and more-detailed survey data on earnings for tipped workers could better assess shortfalls in the tax data and potentially improve the measurement of earnings. Matching worker-level data to employer-level revenue could be another valuable check on tip-reporting in job-level tax filings. Using business data, however, would require making assumptions about the true tip rate, another measure for which more and better quality data are needed.

Our results also have implications for the importance of the tipped-subminimum wage and the tip credit. We find a large share of tipped jobs that seem to pay exactly the tipped-subminimum wage, which implies that this very low statutory wage – $2.13 an hour for jobs in the states subject to the federal minimum – is binding in many instances. In 2022, voters in Washington, DC, passed an initiative that will gradually eliminate the tip credit by 2027. In 2023, the Chicago City Council voted to eliminate the tip credit by 2028. Other states and localities may follow. Given the large share of jobs subject to the tipped-subminimum wage, its elimination could have profound effects on the restaurant industry. In addition to the implications of these laws for workers' earnings, our results suggest that tip-income reporting to the IRS may be affected, because many workers only report enough tips to account for the tip credit: if the tip credit is reduced or eliminated, workers may report fewer tips, although they will likely fully compensate for this reduction with higher base-wage reporting. In general, evaluating the impact of any policy change on the earnings of workers will be difficult if the incentive to report tips is also affected by the change.

Much has happened in the FS restaurant industry in the years since 2018. Automation,

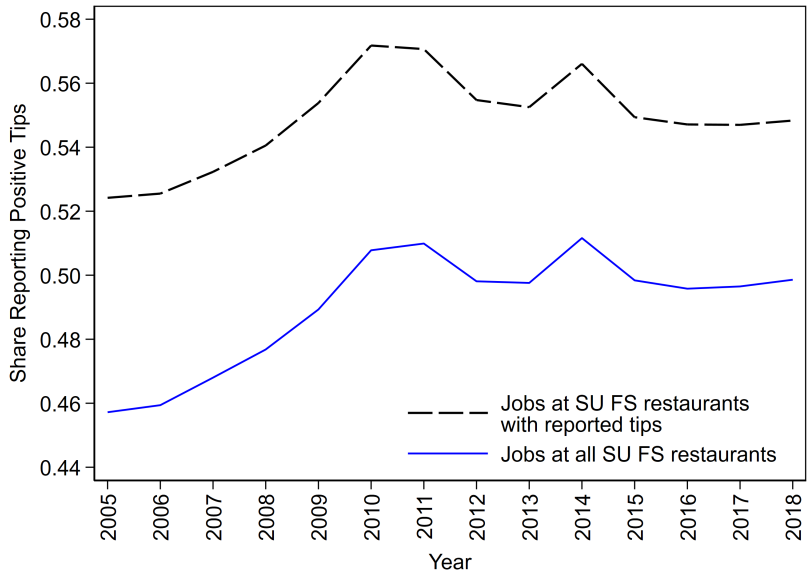
including the use of apps and dedicated devices to view menus, order, and pay, has changed the jobs of servers at many restaurants. Many of these apps and devices include prompts to tip, often at higher rates than patrons had previously applied, and in circumstances – such as ordering take-out – in which many patrons had not previously tipped at all. In addition, apps and devices have increased tipping outside the FS restaurant industry, including in fast-food restaurants. Combined, these changes imply that tipping as a form of compensation will continue to be an important part of earnings for many workers, and, because the apps track tips automatically, administrative records of tips are likely to continue improving at automating restaurants. It remains to be seen whether this effect fully counteracts the reduction in tip reporting in locations where the tip credit is eliminated.

Figure 1. Share of SU FS Restaurants Reporting Tips, 2005–2018



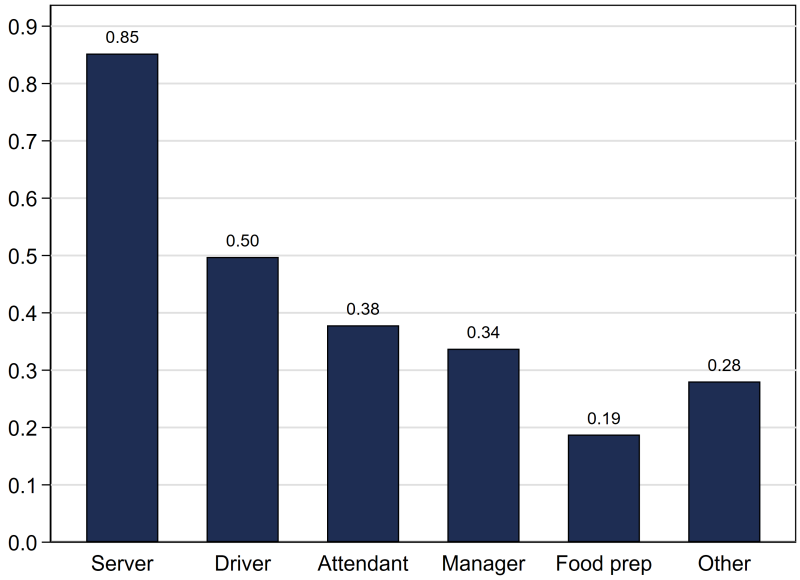
Notes: Authors' calculations from BR and Form W-2 files. Sample of 2,548,000 restaurant-year observations representing 428,000 unique restaurants. Weighted share uses W-2 employment.

Figure 2. Share of Jobs with Reported Tips
at SU FS Restaurants, 2005–2018



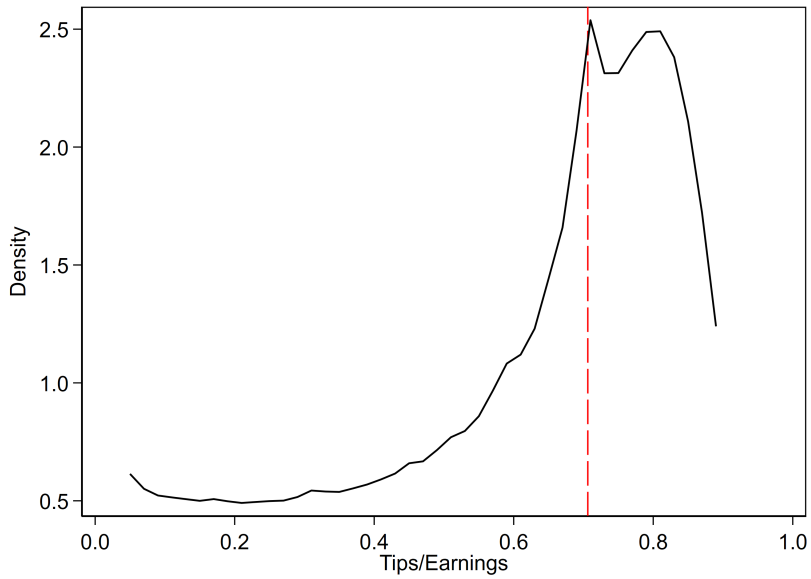
Notes: Authors' calculations from Form W-2. Sample of 68.2 million worker-job-years, representing 21.6 million unique workers.

Figure 3. Share of SU FS Restaurant Workers
Reporting Tips by Occupation, 2005–2018



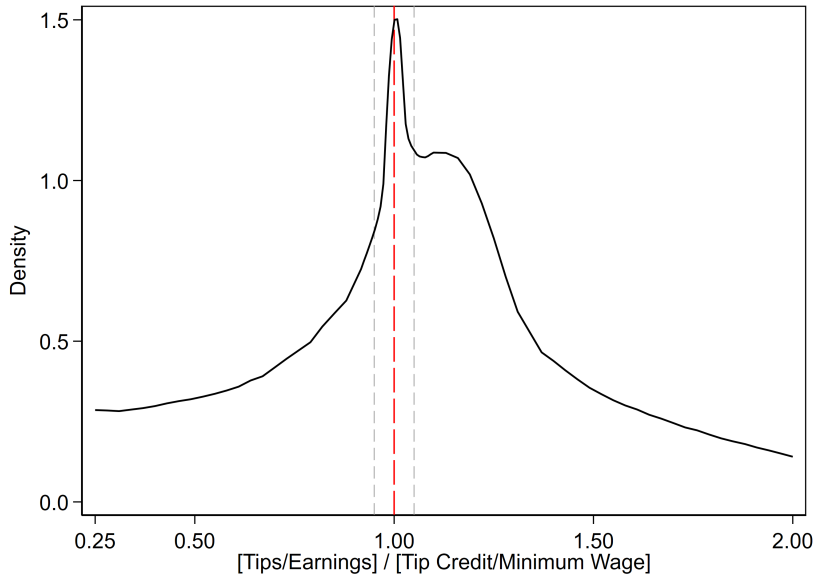
Notes: Authors' calculations from ACS and Form W-2. Sample of 365,000 worker-years who have a SU FS restaurant W-2 and an ACS job classified in the restaurant industry. Occupation is based on either respondents' descriptions of work activities or imputation. Tips are from Form W-2.

Figure 4. Distribution of Reported Tips/Earnings among Tipped Jobs at SU FS Restaurants, States with Federal Tip Credit, 2011–2018



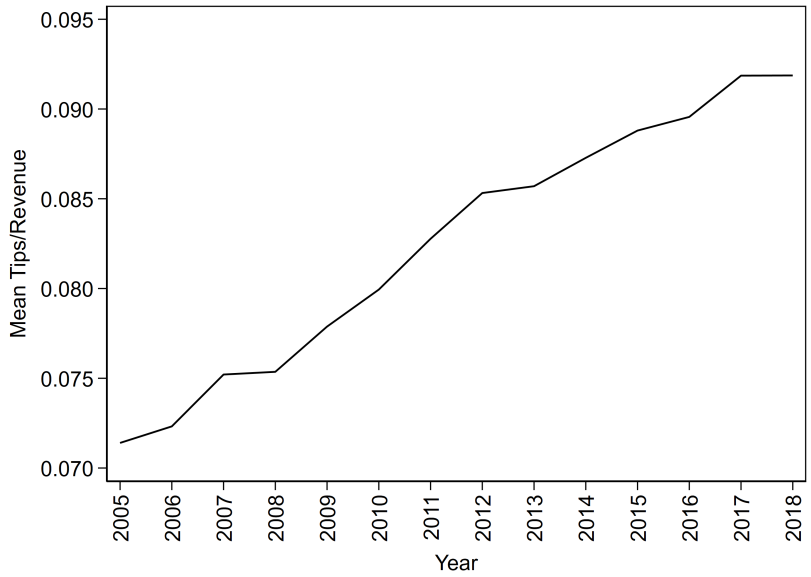
Notes: Authors' calculations from Form W-2 and state minimum wages.
Vertical line at 0.706, the federal tip-credit ratio

Figure 5. Distribution of Reported [Tips/Earnings]/[Tip Credit/Minimum Wage] among Tipped Jobs at SU FS Restaurants Subject to a Tip Credit, 2005–2018



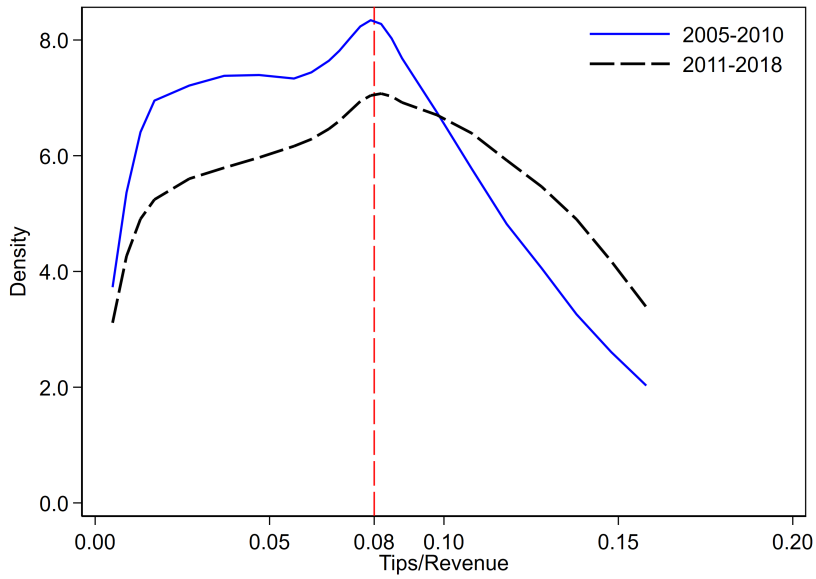
Notes: Authors' calculations from Form W-2 and state minimum wages.
Vertical lines at 0.95, 1.00, and 1.05.

Figure 6. Tips/Revenue at SU FS Restaurants with Tips, 2005–2018



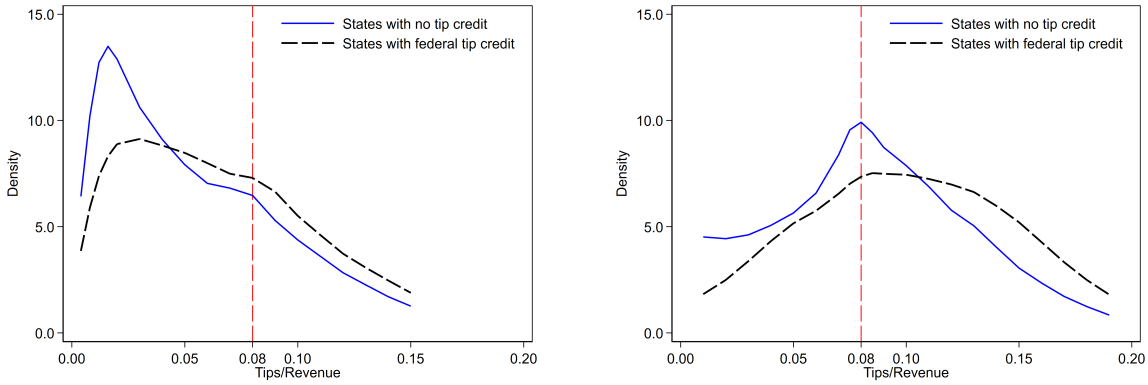
Notes: Authors' calculations from BR and LBD. Sample of approximately 1,500,000 restaurant-year observations. Observations are weighted to adjust for nonrandom missing revenue (see Appendix A.1 for details).

Figure 7. Distribution of Reported Tips/Revenue at SU FS Restaurants with Tips, 2005–2010 and 2011–2018



Notes: Authors' calculations from BR and LBD. Vertical line at 0.08.

Figure 8. Distribution of Reported Tips/Revenue at SU FS Restaurants with Tips, 2011–2018, by Restaurant Size and Tip-Credit Regime

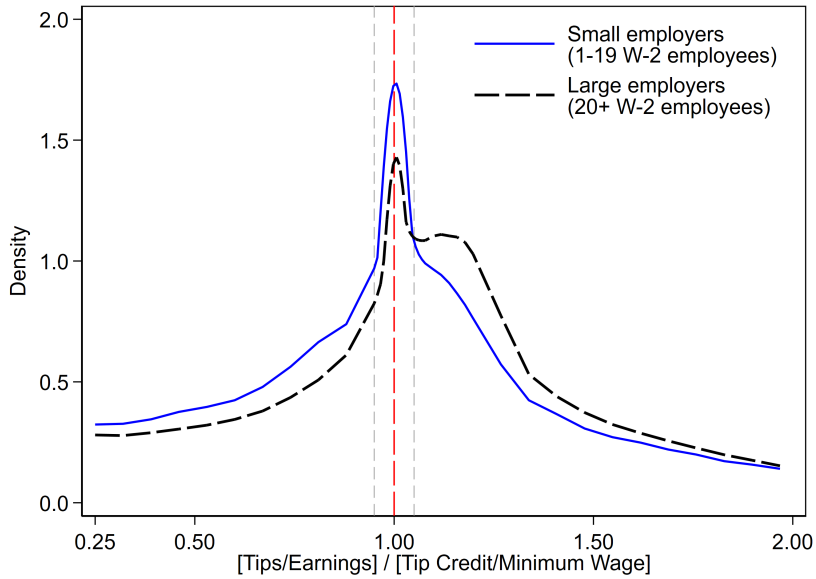


(a) Small Employers (1–19 W-2 Employees)

(b) Large Employers (20+ W-2 Employees)

Notes: Authors' calculations from BR, LBD, and Form W-2.
Vertical line at 0.08.

Figure 9. Distribution of Reported [Tips/Earnings]/[Tip Credit/Minimum Wage] among Tipped Jobs at SU FS Restaurants Subject to a Tip Credit, 2005–2018, by Restaurant Size



Notes: Authors' calculations from Form W-2.
Vertical lines at 0.95, 1.00, and 1.05.

Table 1. Restaurant Size by Tip-Reporting Status, SU FS Restaurants, 2005–2018

	Reported Tips	No Reported Tips
March 12 Employment[a]	19.58	6.319
	(17.01)	(7.325)
Form W-2 Employment[b]	35.25	10.53
	(32.81)	(14.59)
Annual revenue[c,d]	953.1	379.5
	(890.4)	(449.3)
Annual payroll[d]	333.9	80.54
	(363.0)	(126.2)
Reported tips[d]	90.41	
	(115.8)	
Restaurants per year	131,000	51,000

Notes: Authors' calculations from BR, LBD, and Form W-2. Sample of 2,548,000 restaurant-years representing 428,000 unique restaurants. All variables are winsorized at the 99th percentile. Standard deviations in parentheses.

[a] Point-in-time count of workers in the pay period that includes March 12.

[b] Number of workers who worked in the restaurant at some point during the year and received a Form W-2.

[c] Weighted to adjust for nonrandom missing revenue (see Appendix A.1 for details).

[d] Thousands of real 2018 dollars.

Table 2. Reported W-2 Earnings for SU FS Restaurant Jobs, 2005–2018

	Tipped Jobs	Non-Tipped Jobs
Reported earnings[a]	8,986	8,335
	(10,686)	(11,651)
Reported tips	4,727	
	(6,536)	

Notes: Authors' calculations from Form W-2. Sample of 68.2 million worker-job-years, representing 21.6 million unique workers, who have a SU FS restaurant Form W-2 and match to the Numident. Values are in real 2018 dollars and are winsorized at the 99th percentile. Standard deviations in parentheses. "Tipped jobs" are jobs with reported Form W-2 tips. "Non-tipped jobs" are jobs without tips reported on a Form W-2.

[a] For tipped jobs, reported earnings include reported tips

Table 3. Share of Jobs Reporting Tips by Tip-Credit Status and Time Period

	2005–2010	2011–2018
States with No Tip Credit[a]	43.1%	49.3%
States with Federal Tip Credit[b]	46.7%	49.5%

Notes: Authors' calculations from Form W-2 and BR.
[a] States with no tip credit are Alaska, California, Minnesota, Montana, Nevada, Oregon, and Washington.
[b] States in which the federal tip credit applied continuously in 2005–2010 are Alabama, Georgia, Indiana, Kansas, Louisiana, Mississippi, Nebraska, South Carolina, South Dakota, Tennessee, Texas, Utah, Virginia, and Wyoming. States in which the federal tip credit applied continuously in 2011–2018 are Alabama, Georgia, Indiana, Kansas, Kentucky, Louisiana, Mississippi, North Carolina, South Carolina, Tennessee, Texas, Utah, Virginia, and Wyoming. The federal tip credit ranged from approximately 58.6% until July 2007 to approximately 70.6% after July 2009.

Table 4. Tip Reporting in W-2 vs. SIPP, 2013–2016

		Form W-2 tips	
		No	Yes
SIPP Tips	No	0.41	0.25
	Yes	0.05	0.29

Notes: Authors' calculations from SIPP and Form W-2. An observation is a worker-job-year. Sample of 900 worker-job-year observations representing 500 unique SIPP respondents matched with a SU FS restaurant Form W-2.

Table 5. Share of Restaurant Workers Reporting Tip Earnings
by Occupation, CPS-ORG vs. ACS/W-2, 2005–2018

Occupation	CPS-ORG (All)[a]	CPS-ORG (Self Report)[b]	ACS/W-2[c]
Food prep	0.20	0.21	0.12
Attendant	0.12	0.15	0.22
Server	0.60	0.67	0.74
Manager	0.19	0.20	0.20

Notes: Authors' calculations from CPS-ORG, ACS, and Form W-2.
[a] Sample of 102,956 worker-years with a job in the CPS-ORG that was assigned a restaurant industry and occupation based on self-reports, proxy-reports, or imputation. Share usually receives overtime, tips, or commissions as reported to CPS-ORG. Public data.
[b] Sample of 37,212 worker-years with a job in the CPS-ORG that was assigned a restaurant industry and occupation based on self-reports. Share usually receives overtime, tips, or commissions as *self-reported* to CPS-ORG. Public data.
[c] Sample of 811,000 worker-years who have a restaurant Form W-2 and an ACS job assigned a restaurant industry and occupation. Includes workers in multi-unit and SU restaurants, including FS and other restaurant types. Share earning tips as reported on Form W-2.

Table 6. Share of Restaurants Reporting Tips
by Time Period, Restaurant Size, and State Tip-Credit Status

	No Tip Credit[a]		Federal Tip Credit	
Size	2005–2010	2011–2018	2005–2010[b]	2011–2018[c]
All	0.5980	0.6807	0.6835	0.7422
Small[d]	0.4319	0.5133	0.5169	0.5727
Large[e]	0.8384	0.8836	0.8813	0.9111

Notes: Authors' calculations from BR and Form W-2. The unit of observation is a restaurant-year.
[a] States with no tip credit are Alaska, California, Minnesota, Montana, Nevada, Oregon, and Washington.
[b] States in which the federal tip credit applied continuously in 2005–2010 are Alabama, Georgia, Indiana, Kansas, Louisiana, Mississippi, Nebraska, South Carolina, South Dakota, Tennessee, Texas, Utah, Virginia, and Wyoming. The federal tip credit ranged from approximately 58.6% until July 2007 to approximately 70.6% after July 2009.
[c] States in which the federal tip credit applied continuously in 2011–2018 are Alabama, Georgia, Indiana, Kansas, Kentucky, Louisiana, Mississippi, North Carolina, South Carolina, Tennessee, Texas, Utah, Virginia, and Wyoming. The federal tip credit was approximately 70.6% throughout this period.
[d] 1–19 W-2 employees
[e] 20+ W-2 employees

Table 7. Tips/Revenue for Restaurants Reporting Tips by Time Period, Restaurant Size, and State Tip-Credit Status

	No Tip Credit[a]		Federal Tip Credit	
Size	2005–2010	2011–2018	2005–2010[b]	2011–2018[c]
All	0.0652	0.0776	0.0789	0.0944
Small[d]		0.0601		0.0727
Large[e]		0.0899		0.1080

Notes: Authors' calculations from BR and Form W-2. The unit of observation is a restaurant-year. All cells are weighted to adjust for nonrandom missing revenue (see Appendix A.1 for details).
[a] States with no tip credit are Alaska, California, Minnesota, Montana, Nevada, Oregon, and Washington.
[b] States in which the federal tip credit applied continuously in 2005–2010 are Alabama, Georgia, Indiana, Kansas, Louisiana, Mississippi, Nebraska, South Carolina, South Dakota, Tennessee, Texas, Utah, Virginia, and Wyoming. The federal tip credit ranged from approximately 58.6% until July 2007 to approximately 70.6% after July 2009.
[c] States in which the federal tip credit applied continuously in 2011–2018 are Alabama, Georgia, Indiana, Kansas, Kentucky, Louisiana, Mississippi, North Carolina, South Carolina, Tennessee, Texas, Utah, Virginia, and Wyoming. The federal tip credit was approximately 70.6% throughout this period.
[d] 1–19 W-2 employees; reported separately only for 2011–2018
[e] 20+ W-2 employees; reported separately only for 2011–2018

Table 8. Restaurant Size by Tip-Reporting Status, SU FS Restaurants, 2018

	Reported Tips	No Reported Tips
Form W-2 Employment[a]	39.30	11.66
	(36.62)	(17.14)
Annual revenue[b,c]	1,087	471.0
	(945.3)	(540.6)
Reported tips[c]	110.4	
	(132.9)	
Restaurants	150,000	50,000

Notes: Authors' calculations from BR, LBD, and Form W-2. Sample of 200,000 restaurants in 2018. All variables are winsorized at the 99th percentile. Standard deviations in parentheses.
[a] Number of workers who worked in the restaurant at some point during the year and received a Form W-2.
[b] Weighted to adjust for nonrandom missing revenue (see Appendix A.1 for details).
[c] Thousands of real 2018 dollars.

A Data Appendix

A.1 Census Datasets

BR, LBD, and Form W-2

The BR is a list of all business establishments operating in the United States. The LBD tracks businesses across Employer Identification Number (EIN) changes over time; we make use of these longitudinal links whenever possible to mitigate the impact of re-organizations on our linking. For the purposes of this appendix, we refer to the two files as the BR/LBD.

We identify FS restaurants primarily using NAICS codes from the BR. These industry codes incorporate information shared with the Census Bureau by the IRS, SSA, and the Bureau of Labor Statistics but are not based exclusively on administrative data. They are edited by Census Bureau staff to account for responses to various Census Bureau annual surveys as well as the EC. All of these sources rely on businesses to describe their primary activity and are subject to reporting errors. Industry assignment is noisiest for SU firms, the majority of which are "non-mail" cases in the EC, meaning they do not receive any EC forms, so their data come entirely from administrative sources (Basker, 2021). We know very little about the accuracy of industry reporting by firms. In one of the few studies on this topic, Oehlert, Schulz, and Parker (2022, Table 2) use a sample of hand-validated IRS tax returns from 2012–2016 and report an average error rate of about 20% in the reporting of sector (one- or two-digit NAICS). Among businesses that reported sector 72, "Accommodation and Food Services," on their tax forms, between four and eight percent (depending on the form) proved to be in a different sector (Oehlert, 2023, Tables 13 and 17). In our paper, depending on the year, we rely on either the fourth or sixth digit of the industry code to distinguish between FS restaurants and other types of restaurant establishments (limited-

service restaurants, bars, coffee shops, etc.).[42] We have no data on how many errors occur at that level.

BR/LBD payroll, employment, and tips are primarily sourced from IRS Form 941. Businesses file this tax form quarterly to remit FICA payroll and income tax withheld from workers' earnings during the previous quarter and to pay the employer's share of the FICA tax. Payroll is the total amount of wages, tips, and other compensation paid to all employees during the quarter. Similarly, tips are the total amount of tips reported by all employees during the quarter. However, employment is reported at a point in time: workers who received wages, tips, or other compensation in the pay period that includes the 12th day of the third month of the quarter (March 12 (Q1), June 12 (Q2), September 12 (Q3), or December 12 (Q4)). Firms may request permission from the IRS to file an annual Form 944 rather than the quarterly Form 941 if their employment taxes are expected to be below $1,000. In these cases, BR quarterly data are imputed from annual values, but this is relatively rare, even among SU FS restaurants. Prior to 2009, 2–3% of SU FS restaurants filed a Form 944 each year. That share declined in subsequent years; by 2018, fewer than 0.5% filed a Form 944. The BR/LBD treats the 941/944 tax filers as the universe of employer firms in the United States. For more information about the BR/LBD, see Chow, Fort, Goetz, Goldschlag, Lawrence, Perlman, Stinson, and White (2021).

BR/LBD revenue is primarily sourced from annual IRS business-income tax filings on Form 1120, Form 1065, Form 990, and Form 1040 Schedule C, depending on the business's legal form of organization. However, we cannot link all Form 941 tax filers to a revenue tax filing, likely because some firms use different tax identifiers on IRS payroll versus revenue forms. Because of this linking difficulty, revenue information is only available for about 80% of businesses in the LBD (Haltiwanger, Jarmin, Kulick, and Miranda, 2017, p. 17).

[42]For NAICS vintages prior to 2012, FS restaurants were classified as NAICS 7221 and limited-service restaurants were NAICS 7222. Beginning with the 2012 vintage, the NAICS codes were 722511 and 722513 for FS and limited-service, respectively.

Unfortunately, revenue is not missing at random; it is much more likely to be missing for the very smallest of our SU FS restaurants. To address this issue, when we calculate average revenue per restaurant in Tables 1, 7, and 8, we use propensity-score weights to adjust for the nonrandom pattern of data. Specifically, we first estimate a probit model predicting the availability of revenue information using the log number of Form W-2 workers, log annual payroll, indicators for establishment birth and death years, and calendar-year and state fixed effects; we use the inverse predicted probability to weight observations with non-missing revenue. Figures 7 and 8 are unweighted. For more information about revenue in the LBD, see Haltiwanger, Jarmin, Kulick, Miranda, Penciakova, and Tello-Trillo (2019) and Yeh and Macaluso (2023).

Form W-2 data are available by worker-job-year, where a job is defined as a match between a business and a person. These individual tax forms are filed by employers for all jobs for which FICA taxes are withheld, which includes virtually all restaurant jobs.[43] We drop a small number of workers whose identifiers are associated with more than 24 unique employers in a given year. The IRS requests that Form W-2 and Form 941 be filed using the same business tax identifier (EIN) so that IRS can ensure that the amounts match across the two forms.[44] Because of this, we can match most individual Form W-2 records to BR/LBD restaurant records. However, there are cases in which we do not match the two sources, as seen in Table A-1. These could be the result of late or missing filings or change in EIN due to change in ownership. In our analysis of restaurant-level reporting, we treat any reporting of tips – whether on a Form W-2 for a worker at the business, or on a Form 941 for the business as a whole – as evidence that the business is reporting tips to the IRS. Our analysis of extensive-margin misreporting therefore focuses on the 28% of SU FS restaurants that do not match to any tip reporting in our records.

[43]See `https://www.irs.gov/pub/irs-prior/iw2w3--2005.pdf` and `https://www.irs.gov/pub/irs-prior/iw2w3--2018.pdf`, accessed May 13, 2024.

[44]See `https://www.irs.gov/instructions/i941#en_US_202403_publink1000294661`, last accessed April 30, 2024.

IRS rules require workers to report tip income to their employers, which are then responsible for entering the information on the workers' Form W-2, and on Form 941 quarterly filings at the business level. Tips are reported separately on Form W-2 because they are treated differently from wage earnings for the purpose of FICA taxes.[45] Workers pay half of the payroll taxes on their tips and employers pay the other half but are allowed to take a credit against their income tax for payroll taxes on tips that contribute to workers earning more than $5.15 an hour.[46] Because of this employer rule, wages subject to FICA tax are reported in Box 3 (Social Security wages) and tips subject to FICA tax are reported in Box 7 (Social Security tips). In practice, Form W-2 data housed at the Census Bureau do not include tips reported to the IRS, so we infer tips by subtracting Form W-2 Box 3 income (Social Security wages) from total compensation, which includes Box 1 income (wages, tips, other compensation) and Box 12 income (deferred compensation, such as 401(k) and Health Savings Account contributions, and other compensation, such as moving expenses). This difference gives us the portion of total compensation not treated as Social Security wages and is a valid measure of tips for employees of restaurants as long as the worker's total earnings are less than the FICA taxable maximum, which varies by year. In 2018, the FICA maximum was $128,400, well above what most tipped employees earn from tips and wages combined.[47] We drop any worker-job-year observation if Box 3 income is at that year's FICA maximum to avoid over-imputing tips to non-tipped jobs. In addition, our method of calculating tips combines regular tips (Box 7) and allocated tips, which are required when a restaurant reports insufficient tips to meet its required revenue share (Box 8).[48] However, IRS tabulations for 2008–2018 show that allocated tips are well under 1% of total reported

[45]The federal payroll tax, often referred to as FICA based on the law that authorized it, funds the Social Security Old-Age, Survivor, and Disability Insurance (OASDI) program and Medicare.

[46]See U.S. Department of Treasury, Office of Tax Analysis (2016) for more details.

[47]For a history of FICA taxable maximum amounts see `https://www.ssa.gov/oact/cola/cbb.html`, last accessed April 30, 2024.

[48]As noted in Section 2 and throughout the paper, that threshold is typically 8%. However, restaurants may petition the IRS for a lower rate, which can be as low as 2%. (See `https://www.irs.gov/instructions/i8027`, last accessed October 23, 2024.) We do not know how often such petitions are made or granted.

tips every year (Internal Revenue Service, 2017). This is consistent with our finding that most restaurants meet the 8% revenue-share threshold for tips if it applies to them, so allocating tips is generally not required.

Economic Census

The Economic Census (EC) is conducted quinquennially (every five years) in years ending in "2" and "7." We use data from the 2007, 2012, and 2017 ECs. Due to concerns about respondent burden, some very small restaurants do not receive an Economic Census form (administrative cases). Of those that do, some do not respond at all (survey non-responders) and some may not answer specific questions (item non-responders).

For restaurants that do respond, one question the EC asks is average meal price, where price is selected from a checklist and is intended to capture the estimated average price per meal, excluding tax and tip.

Table A-2 shows the mean ratio of tips to revenue by nominal price bin in 2012 and 2017. It shows that reported tip rates are higher for more expensive restaurants, which may reflect the fact that patrons actually tip more at more expensive restaurants or be a feature of differential reporting. Cheaper restaurants may be those that rely on more takeout, for example, and may be less likely to serve alcohol, both of which could affect actual tipping. But they may also be less formally managed and have poorer record keeping.

A second observation is that for the higher-priced restaurants (average meal price of $10 or more), the ratio of tips to revenue increased between 2012 and 2017, consistent with the increasing trend of tips to revenue over the full sample period, seen in Figure 6.

Worker Demographics

We obtain demographic information – age and sex – from the Census Bureau's version of the Numident, which is a processed version of the SSA Numident, stripped of Social Security Numbers, but containing a longitudinally consistent Protected Identification Key (PIK) for

each individual that allows us to link it to other datasets. For more information about the Numident, see Finlay and Genadek (2021).

The demographic variables allow us to zoom in on the sample of servers from Figure 3 to check tip-reporting rates by sex and age. Table A-3 shows the results. The first two rows show the breakdown by sex: 86% of female servers and 84% of male servers reported tips on one or more of their Form W-2s. The next three rows show a breakdown by worker age: 73% of teenagers, 88% of workers in their 20s, and 87% of workers in their 30s and older report some tips to the IRS. Overall, tip-reporting rates seem very stable across demographic groups, with the possible exception of lower reporting by teenagers.

Teenagers' lower reporting rates could be due to factors other than under-reporting of tips, such as higher rates of occupation misreporting or different task assignment within the occupation, which could be related to working at different types of restaurants. We confirm this using tip reporting by age group in the CPS-ORG. Restricting to workers who report a server occupation in the CPS-ORG, only 45% of workers ages 15-19 report earning any tips, overtime or commission, compared to 65% of workers in their 20s and 61% of workers ages 30 and over. This finding suggests that occupations based on self-described work activities do not fully capture differences in jobs across age groups.

Unemployment Insurance

One more source of information at the job level is the Longitudinal Employer-Household Dynamics (LEHD) data infrastructure, which captures state UI records. Most Form W-2 jobs can be matched by EIN and PIK to state UI records. In some cases, however, employers may use different tax identifiers for reporting employment to their state UI systems and for reporting to the IRS, in which case we are unable to match earnings records across sources. (Goldschlag, Jarmin, Lane, and Zolas, 2021, Appendix IA.2, report that, across the universe of Form W-2 records in 2005–2014, an LEHD record is missing in approximately 20% of cases; conversely, Form W-2 is missing for approximately 10% of LEHD observations.) Because

LEHD includes Massachusetts only starting in 2010 (Hyatt, McEntarfer, McKinney, Tibbets, and Walton, 2014), we exclude restaurants in Massachusetts from the analysis in this paper. For more information about the LEHD data infrastructure, see Abowd, Stephens, Vilhuber, Andersson, McKinney, Roemer, and Woodcock (2009) and Vilhuber (2018).

LEHD data include total earnings by worker-job-quarter, and therefore allow us to identify how many quarters a job lasted. We use this information for a few purposes.

First, as detailed in Section A.2, minimum wages apply at the business level. However, minimum wages sometimes change during a calendar year, as when the federal minimum wage increased in July of 2007, 2008, and 2009. Because many jobs in the restaurant industry last less than a full calendar year, we use LEHD information to more precisely identify the effective minimum wage at the job level for Figures 5 and 9. For jobs that do not match to LEHD, we use the mean value of the minimum wage faced by the business in a calendar year.

Second, we use LEHD to calculate the average number of quarters in 2018 that a job lasted. We do this to translate the dollar value of missing tips per job in 2018 to a dollar value per job-quarter.

Finally, we use LEHD to calculate the number of quarters for each job in the 2018 CPS-ASEC/NEWS sample. This requires matching Form W-2 records to UI records by EIN and PIK. For non-matched jobs, we assign the average quarters worked at a job.

Survey of Income and Program Participation

We use the 2014 SIPP panel, which covers the period 2013–2016. This panel includes approximately 34,000 unique individuals with jobs, many of whom have multiple jobs in a year. The SIPP allows workers to report up to seven jobs in a year, and includes information on each employer's name and address. We use the name and address information to find the employer in the BR, and use the employer's EIN to identify the worker's Form W-2. We

match 900 worker-job-year observations in the SIPP, representing 500 unique workers, to SU FS restaurant employers.

Occupations in Survey Data

Approximately 640,000 observations in the ACS over the period 2005–2018 can be linked via Form W-2 to jobs in SU FS restaurants in the year in which they were included in the survey. Each of these workers had an industry and occupation code assigned, based either on a description of the primary job provided by the respondent or an imputation in the case of missing data. Because workers may have multiple jobs in a year, both simultaneously and sequentially, and because restaurant jobs are often "second jobs," only 365,000, or about 57%, have their primary industry coded as the restaurant industry. Our ACS analysis in Figure 3 uses these 365,000 respondents who had both a SU FS restaurant Form W-2 and a corresponding ACS restaurant job report.

In the analysis in Table 5, we include a larger set of workers: any worker in the ACS or CPS-ORG who was assigned a restaurant industry and occupation, either using self-reports about work activity or imputation. This sample of about 811,000 ACS respondents and a little over 100,000 CPS-ORG respondents includes workers at all restaurants, not just FS restaurants, and therefore has occupations, such as counter attendants, that are unlikely to be found at FS restaurants.

Finally, we use occupation information in the SIPP for jobs we match to a Form W-2 that also reports tip earnings, and for which the worker also reports tips to the SIPP. For this sample of approximately 260 worker-job-years, we restrict our attention to jobs described as having a server or food-prep occupation.

The survey occupation categories are very detailed, and we group them into several occupational groups. Specifically, we include chefs, cooks, and workers in food-preparation occupations, as well as combined food-preparation and serving workers, under "food prep"; counter attendants, dining-room and cafeteria attendants, hosts and hostesses, dishwashers,

and cashiers, under "attendant"; bartenders, waiters, and waitresses under "server"; and first-line managers and food-service managers under "manager." All occupations other than these, with the exception of drivers, are classified as "other."

Occupation codes in household surveys are known to have measurement error.[49] Abraham and Spletzer (2009) compare the CPS-ORG to information collected by the Occupational Employment Statistics survey, a large employer survey, and find evidence consistent with significant misclassification, specifically with exaggerated occupational status in the survey. In our context, such exaggeration could correspond, for example, to a shift supervisor reporting a management occupation, or a busser reporting a server occupation. Houseworth and Fisher (2020, Table 1) summarize misreporting rates in the literature as falling between 6.5–24.3% for one-digit occupation and 42.6–57.6% for three-digit occupation. Almost all the occupations we include in this paper have the same first digit (the exceptions are drivers and some managers), leading us to expect higher levels of occupation misreporting. Houseworth and Fisher (2020) also find that misreporting is higher for proxy responses, although the magnitude of that difference is small relative to baseline misreporting rates.

A.2 Other Datasets

Minimum Wages

We use monthly minimum-wage data at the state and local level from 2005 to 2018. We deflate each monthly value using the all-items CPI, and then average by calendar year.

Our starting point for the regular minimum wages is a monthly file from Basker and Khan (2016) with state-level minimum wages for the period 1993–2014. We extend the dataset by four years using the Department of Labor's (DOL) website.[50] Our starting point

[49] There may also be errors in industry codes.

[50] See `https://www.dol.gov/agencies/whd/state/minimum-wage/history`, last accessed March 11, 2021.

for the tipped, subminimum, wages is an annual file created by Jones (2016). We supplement this file using another DOL website to create monthly records.[51]

For both the minimum and subminimum wages, we supplement state-level files with information on city- and county-specific minimum-wage ordinances. In some cases these ordinances were repealed or litigated and annulled by the courts. We have done extensive searches in city codes and news reports to compile the start and, when applicable, end dates of these higher minimum wages. In all, we have found approximately 45 localities that imposed a higher minimum wage than their states at some point during the sample period.

Some states and localities have different minimum wages for different types of jobs or businesses. For example, New York has different minimum wages for different industries and occupations; we use the minimum wage for food-service jobs in the restaurant industry. Connecticut has two different tipped-subminimum wages: one for bartenders, and one for all other waitstaff. We use the waitstaff subminimum wage for our tip-credit calculations. West Virginia instituted a lower minimum wage for employers with five or fewer employees, starting in July 2006, and Ohio instituted two different minimum wages, one for businesses with under $250,000 in revenues and one for those those with more than $250,000 in annual revenue, in 2007; the revenue threshold has moved annually since then. In these and similar cases, we use information on employment and revenue from the BR/LBD to assign the relevant minimum wages to each restaurant.

Although Washington bans tip credits, Seattle instituted a different minimum wage for tipped jobs starting in 2015 in restaurants belonging to chains with up to 500 employees worldwide.[52] These jobs must earn tips but there is no dollar amount that the employer needs to make up for the jobs to be eligible for the lower minimum wage. In Hawaii, also starting in 2015, tipped employees must earn at least $7/hour in tips to be eligible for a

[51] See `https://www.dol.gov/agencies/whd/state/minimum-wage/tipped/History`, last accessed March 11, 2021. Even and Macpherson (2014) construct a similar dataset ending in 2013.

[52] See `http://www.seattle.gov/Documents/Departments/LaborStandards/OLS-MW-multiyearChart2019FINAL10118(1).pdf`, accessed March 25, 2021.

$0.50–0.75 tip credit.[53] The District of Columbia passed a ballot initiative in 2022 that will eventually eliminate the tip credit (D.C. Law 24-281), and the Chicago City Council passed a similar measure in 2023 (Romeo, 2023).

Finally, whenever the federal minimum wage exceeds a state or locality's statutory minimum wage, or the state has no statutory minimum wage, the federal value applies.

FoodAPS

Just as it is rare to find non-tax data on tip earnings by workers, it is rare to find data on tipping by customers. We do not use the FoodAPS data in this paper, but in this section, we compare tip reporting by customers to tip reporting by restaurants.

Between April 2012 and January 2013, the Economic Research Service (ERS) at the U.S. Department of Agriculture (USDA) conducted the National Household Food Acquisition and Purchase Survey (FoodAPS), which surveyed households about their food consumption and spending. The survey targeted three types of households: households that received Supplementary Nutritional Assistance Program (SNAP) benefits, households below the poverty line that did not receive SNAP benefits, and households well above the poverty line.[54]

Among many other details, for meals consumed away from home, the survey asks about the type of location (e.g., another person's home, school, vending machine, restaurant) as well as the number of people who participated in the meal. For restaurants, there is further information about the total amount paid (including tax and tip), the tip amount, and whether the restaurant is one of the top 30 fast-food chains or top 30 casual-dining restaurants.[55]

[53] See `https://labor.hawaii.gov/wsd/files/2014/06/Tip-Credit-Notice-with-exhibits-June-2014.pdf`, accessed September 26, 2021. Because the Hawaii tip credit is not equal to the difference between the minimum and subminimum wages, we omit Hawaii from Figures 5 and 9.

[54] The public-use data files from this survey are available at `https://www.ers.usda.gov/foodaps`; accessed August 24, 2023.

[55] The top 30 fast-food chains, as coded in this variable, include Arby's, McDonald's, Starbucks, and Panda Express. The top 30 causal-dining chains include Olive Garden, Denny's, Waffle House, and P.F. Chang's. A full list is available in the FoodAPS documentation.

To maximize the comparability of the survey sample to our EC sample of SU FS restaurants, we drop FoodAPS meals at the top-30 fast-food chains and the top-30 casual-dining chains. We further restrict to meals with non-missing (not necessarily nonzero) tips, and for which the total amount paid is positive and exceeds the tip (because the tip is included in the total amount paid), dropping meals that are coded as "free." This leaves us with a total of 1,992 reported meals, almost all of which took place in 2012. We calculate the tip percentage as the ratio of the tip to the pre-tip total meal cost. The average ratio of tips to meal cost is 14.6%.

Square

We use average tip rates by state in 2014 as published by the Square payment application (Quartz, 2014). Square provides a mechanism for businesses to process payment-card transactions without a traditional credit-card reader. There is no specific time period provided for these rates, so we assume they apply to 2013–2014. Square was primarily used by small businesses during this time: in 2012, more than 75% of Square's payment volume came from merchants with under $125,000 in charges; in 2014, as the application become more established, that number declined to 60% (Square, 2014). Flynn (2014) estimates that Square was used by 50,000 restaurants during this period.

Quartz (2014) reports separate numbers for restaurants, food trucks, and coffee/tea shops; we use only the restaurant figures in our calculations. On average across states, approximately 56.8% of all restaurant transactions on Square included a tip. As noted by Quartz (2014), this does not necessarily mean the remaining transactions did not tip: some likely left a cash tip. Among tipped transactions, the average tip rate was approximately 16.5%.

Table A-1. Tip Reporting by Source, SU FS Restaurant Years

		Form W-2 tips: No	Form W-2 tips: Yes
Form 941 Tips	No	0.28	0.02
	Yes	0.01	0.69

Notes: Authors' calculations from BR and Form W-2. Sample of 2,548,000 restaurant-year observations representing 428,000 unique restaurants. Form W-2 reporting is aggregated across all jobs at a given employer-year.

Table A-2. Tip/Revenue by Meal Price, SU FS Restaurants, 2012 and 2017

Average Meal Price[a]	2012[b]	2017[c]
Under $7	0.078	0.045
$7.00–$9.99	0.092	0.082
$10.00–$14.99	0.103	0.108
$15.00–$19.99	0.114	0.121
$20.00–$29.99	0.124	0.128
$30.00 or more	0.141	0.146

Notes: Authors' calculations from EC and BR/LBD. Average meal price and revenue are from the EC; tips are from BR. Samples include only restaurants with EC revenue and reported Form 941 tips.
[a] Meal price is nominal
[b] Sample of 34,500 restaurants
[c] Sample of 22,000 restaurants

Table A-3. Share of SU FS Restaurant Servers Reporting Tips by Demographics, 2005–2018

Demographic Group	Share Tipped
Female	0.86
Male	0.84
Age 15-19	0.73
Age 20-29	0.88
Age 30+	0.87

Notes: Authors' calculations from ACS and Form W-2. Sample of 150,000 worker-years who have a SU FS restaurant Form W-2, and an ACS job with a restaurant industry classification and server occupation classification. Tips are from Form W-2.

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